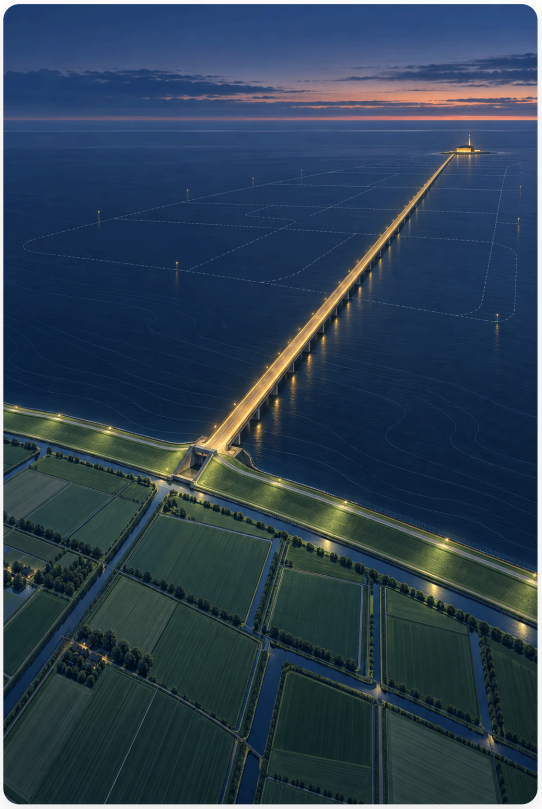


The Netherlands: The Feasibility Debt

Commitment Before Durable Feasibility

The Netherlands is not failing. It has built one of the most sophisticated systems in the world for governing an uncertain future, above all in water. Yet outside its mature water architecture, it has repeatedly committed to futures before securing them: nitrogen permits resting on ecological gains that had not yet occurred, renewable subsidies awarded before grid capacity was checked, and a national airport operating for roughly a decade ahead of the legal regime meant to govern it. This report diagnoses the Feasibility Debt and proposes a Feasibility Balance Sheet as the concrete first step.

AI-generated · prompted and curated by Björn Kenneth Holmström

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Country Report · Netherlands

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Executive Summary

The Paradox

No country has more experience of living inside hard limits than the Netherlands. Nearly 60 percent of its territory could be exposed to flooding. After the storm surge of 1953, it built the Delta Works, and over the following decades it developed something rarer than great engineering: a way of governing a threat whose future scale cannot be known. Adaptive delta management—with its tipping points, early decision points, alternative pathways and periodic revalidation—is now studied around the world as a model for planning under deep uncertainty.

And yet. On the reclaimed land of Flevoland stands Lelystad Airport, its runway extended, its control tower enlarged and its terminal completed for commercial flights planned for April 2018. No commercial flight has yet departed. For roughly a decade, Schiphol operated under a formally temporary arrangement in which the state declined to enforce legally binding noise limits while it waited for a replacement regime to be written into law. Around 700 megawatts of subsidised renewable projects were approved before anyone had checked whether the grid could carry their electricity. And for four years, nitrogen permits were granted partly on the strength of ecological improvements that had not yet occurred—until, in 2019, the Council of State ruled that such future gains could not be spent in advance.

These are not the failures of a country that ignores constraints. They are the failures of a country so accomplished at engineering its way through constraints that, outside its mature water architecture, it has sometimes committed to a future before securing it.

The Core Diagnosis: Feasibility Debt—Commitment Before Durable Feasibility

Feasibility debt accumulates when durable commitments are made before the ecological, infrastructural, fiscal, administrative or legal capacity needed to execute them has been secured across the relevant implementation horizon. Its defining feature is simple: **commitment precedes durable feasibility.**

Feasibility is not a moment but a path. A project can be feasible on the day it is approved and infeasible by the day it is built, because other projects consume the same capacity, the law changes, or the ecological assumptions on which it rested prove wrong. The Dutch evidence shows three forms of the debt:

- **Borrowed future capacity.** Capacity expected in the future is treated as available today. The Programmatic Approach to Nitrogen (PAS) authorised emissions partly against ecological gains its future measures were expected to deliver; under European nature law, which demands certainty that protected habitats will not be harmed, that expectation was not enough.
- **Unsecured implementation capacity.** A commitment is made before the capacity needed to carry it out is secured. Renewable subsidies were awarded without project-specific grid checks until 2019; Lelystad's infrastructure was completed before its airspace, nature permit and operating decree were secured.

- **Anticipated legalisation.** Current operations rely on a legal regime expected but not yet in force. Schiphol's anticipatory enforcement, begun in 2015 as a bridge, had not been fully retired by September 2026.

This report does not claim that feasibility debt explains most of what goes wrong in Dutch governance; the evidence establishes recurrence across distinct domains, not dominance. It does not claim that the decisions involved were irrational. And it does not claim that the courts caused the crises in which the debts came due. The ecological, physical and legal boundaries existed before any judge enforced them.

The Twin Deficits

Aspect	Outer (Hardware)	Inner (Operating System)
Strength	The Delta architecture—Delta Act, Commissioner, Fund and adaptive pathways; explicit allocation systems for freshwater, grid capacity and nitrogen space; independent courts; a capable, largely incorruptible administration	*Maakbaarheid*: the confidence, earned over centuries, that the country can shape its own environment; the polder tradition of negotiated, broadly owned settlements
Deficit	Commitments made before feasibility in nitrogen, energy and aviation; the Delta package of continuity, integration, finance and pathway switching present only in pieces outside water; a rural transition architecture designed, funded and abandoned	*Maakbaarheid* in its less disciplined form—the assumption that feasibility can be made later; consensus culture's tendency to postpone the allocation of permanent losses; a political culture comfortable with a managed gap between rule and enforcement
Manifestation	A finished terminal at Lelystad with no commercial flights; subsidised projects without grid connections; PAS reporters left without valid permits; a decade of tolerated non-compliance at Schiphol; an infrastructure programme paused because it exceeded its combined envelope of money, permits and staff	Farmer protests and the rise of a new rural party; a failed agricultural agreement; residents living with noise beyond legal limits; "the country is full" as a popular diagnosis that misses the point

The Signature Pattern: The Pressure–Commit–Bind–Repay Loop

Each report in this series identifies a characteristic motion. Japan's architecture preserves; Korea's competes; Bhutan's races. The Dutch motion is a cycle of borrowing and repayment.

Pressure. Simultaneous, legitimate demands—around 100,000 homes a year, rapid electrification, a hub airport, one of the world's most productive agricultural sectors—press against a densely occupied physical, ecological and legal environment.

Commit. Under pressure, commitments are made on capacity that is assumed: ecological improvement expected, grid capacity to be built, a legal regime to be enacted.

Bind. The real boundary asserts itself, enforced by a court, by physics, by finance or by ecology.

Repay. The debt is paid in delay, redesign, compensation, restriction or cancellation—and the conflict over who pays feeds the next round of pressure.

How the Debt Is Called In—and Why Some Debts Are Harder to Repay

Courts have repeatedly called in feasibility debt: the 2019 PAS ruling; the 2024 tightening of internal nitrogen offsetting; the 2025 order to meet the statutory 2030 nitrogen target; and, in climate, the 2019 *Urgenda* judgment. They acted as intertemporal constraint enforcers, refusing to accept uncertain future mitigation as a substitute for present compliance. But debt does not require a court. On the grid, physics called it in. In the national infrastructure programme, seventeen projects were paused in 2023 because the programme exceeded its combined envelope of finance, nitrogen permit space and delivery capacity.

Three properties of the underlying constraint explain much of why some debts are repaid through a queue and others through a political crisis: the **certainty** of the missing capacity, the **permanence** of the losses repayment requires, and the **spatial specificity** of the constraint. Grid debt involves relatively certain, expandable capacity and mostly temporary losses; it is painful but tractable. Nitrogen debt involves uncertain capacity, permanent losses and capacity tied to specific protected habitats; it has convulsed Dutch politics. Water provides the within-domain test: temporary drought rationing works through an established priority sequence, while the permanent land-use change the Delta Programme now says is necessary is proving much harder.

The Legitimacy of Permanent Loss

The hardest debts require someone to give something up permanently, and that makes legitimacy part of the constraint set. The rural transition that might have given nitrogen a Delta-like architecture—long-horizon area programmes backed by a dedicated fund—was designed, funded and then abandoned amid farmer protests, a failed agricultural agreement and a change of government. Yet when compensated exit was offered on acceptable terms, demand could be strong: one cessation scheme was oversubscribed within a day. The problem is not that permanent loss can never be negotiated. It is that it requires an allocation process that those affected regard as fair, adequately funded and stable enough to plan around.

The Series Boundary Condition

Most countries in this series suffer first-order deficits; others meet structural ceilings. The Netherlands resembles Bhutan and Finland in facing a problem of timing rather than incapacity. In Bhutan, the risk is that interfaces lag behind accelerating institutions. In Finland, a competent system cannot transform quickly

enough. In the Netherlands, commitments have outrun feasibility—and the country already holds, in its water governance, one of the most developed remedies any state has produced. Its challenge is not invention but transfer.

The Opportunity

The counter-model is **recursive feasibility**: governance that preserves at least one credible feasible path from commitment through operation. The Netherlands already practises it for water. Extending it means applying the Delta method's functions, not copying its institutions:

- a **feasibility-pathway assessment** for major national commitments, answering nine questions before commitment—including, crucially, who bears permanent losses if the debt must be repaid;
- a four-rung ladder of protection—**observe, forecast, conditionally preserve, revalidate and release**—with the grid's "use it on time or lose it" rules as proof that preservation without release creates its own scarcity;
- **decision points set before tipping points**, allowing for the lead time alternatives require;
- continuity, independent integration, long-horizon finance and pathway switching, transferred by function to the domains that lack them;
- a **legitimacy architecture for permanent loss**, drawing on Room for the River, the national programme that gave land back to the rivers and showed that Dutch governance can allocate permanent, place-specific losses when compensation is fair, involvement early and the programme stable.

The Trojan Horse is the Delta method itself: *Delta-proof decisions*, the application of the country's most trusted governance tradition to its other long-horizon commitments.

The Concrete First Step: The Feasibility Balance Sheet

The Netherlands accounts meticulously for its fiscal debt. It has no account of the commitments it has made against capacity it has not secured. A **Feasibility Balance Sheet**, compiled by an independent institution and presented to Parliament on Budget Day alongside the fiscal accounts, would record each outstanding debt—its form, scale, exposure, likely enforcer and repayment pathway—together with contingent debts: commitments that depend on future delivery that is not yet secure. Paired with feasibility-pathway assessments for new commitments, it would not prevent the government from borrowing against the future. It would prevent it from doing so invisibly. Four registered predictions—on water quality after 2027, the new housing locations, energy infrastructure and nitrogen governance—would test whether the diagnosis holds.

The Honest Conclusion

Feasibility debt recurs across ecological, infrastructural and legal domains in the Netherlands, and its repayment is hardest where losses are permanent and tied to place. The country has already begun to test feasibility before committing in several domains, though the outcomes are not yet proven. Whether the Delta

method's functions can be carried into domains with different authorities, laws and losses—and whether Dutch politics can sustain the arrangements needed to allocate permanent loss—remains open.

The transition is feasible. The Netherlands has already demonstrated, at home, every capability it requires. What it has not yet demonstrated is the will and the continuity to apply them beyond the domain where they were born.

Series Coherence Table (Updated)

System	Core Deficit	Signature Pattern	Cultural Anchor	Transition Feasibility
Germany	Execution	Paralysed spending	Engineering rigour	Feasible
France	Integration	Reform-explosion-retreat	Jacobin clarity	Feasible
Sweden	Feedback	Drift loop (signal suppression)	*Saklighet*	Feasible
India	Synchronisation	Leap-lag cycle	*Jugaad*	Feasible
EU	Coherence	Negotiation-dilution	Subsidiarity	Feasible
UK	Control-delivery mismatch	Centralise-fail-centralise	Muddling through	Feasible
Brazil	Accumulation	Breakthrough-Capture	*Jeitinho*	Difficult but possible
Russia	Legibility	Control–Blindness–Shock	*Ne vysovyvaysya*	Impossible under current regime
USA	Integration	Escalate-Block-Bypass-Delegitimise	Bootstrap individualism	Possible via sub-federal
Finland	Throughput Constraint	Anticipate–Consensus–Increment–Pressure	*Sisu* + Quiet Consensus	Feasible
China	Calibration	Campaign–Overshoot–Abrupt Correction	*Míng zhé bǎo shēn*	Difficult; recoverable under current regime
Japan	Continuity Trap (Paradigm Lock-in)	Pressure–Accommodate–Preserve–Defer	*Wa* + *Kaizen* + *Gaman* + *Shouganai*	Feasible with controlled creative destruction
Nigeria	Substrate Deficit (State–Society Dissociation)	Extraction–Dissociation–Adaptation–Crisis	*Oga-Madam* + "The National Cake" + *Jugaad* + Pentecostal Resilience	Generational; feasible via interface-building from below

System	Core Deficit	Signature Pattern	Cultural Anchor	Transition Feasibility
Israel	Boundary Deficit (Contingency Lock-In)	Threat–Mobilization–Securitization–Fragmentation–Renewed Threat	*Ein Breira* + *Balagan* + Covenant Consciousness + *Tikun Olam*	Difficult; requires constitutional settlement via incremental boundary stabilization
Spain	Integrative Closure Deficit (Transition Trap)	Crisis–Centralisation–Peripheral Mobilisation–EU Mediation–Accommodation	*Convivencia* + *Las Dos Españas* + *El Aplazamiento*	Feasible via orthogonal interventions; unlikely via direct constitutional reform
South Korea	Tournament Trap (Consequence Coupling Under Effective Scarcity)	Effective Scarcity–Rank–Invest–Concentrate	*Gwageo* + *Hakbeol* + *Gongjeong*	Feasible if stakes are lowered before selection is changed
Bhutan	Interface Race (Objective–Actuator Coupling Under Acceleration)	Compare–Exit–Accelerate–Recouple	GNH + *Kidu* + *Driglam Namzha* + *Tha Damtshig Ley Jumdrey*	Transition case; outcome depends on interface construction keeping pace
Netherlands	**Feasibility Debt (Commitment Before Durable Feasibility)**	**Pressure–Commit–Bind–Repay**	***Maakbaarheid* + *Polder Negotiation* + *Bounded Resonance with *Gedogen***	**Feasible — domestic counter-model exists; transfer across domains remains unproven**
The Netherlands is the case that asks: *What*				

System	Core Deficit	Signature Pattern	Cultural Anchor	Transition Feasibility
happens when the country that taught the world to plan for an uncertain future sometimes commits to its own before securing it? The answer lies not in invention but in transfer: carrying the wisdom it developed at sea into the decisions it now makes on land.				

1. The Feasibility Debt

1.1 Opening: The Airport Without Flights

On the reclaimed land of Flevoland, some fifty kilometres east of Amsterdam, stands an airport that is ready and has never opened. Its runway was lengthened and widened for commercial jets. Its control tower was enlarged. Its terminal was completed. Everything was prepared for the first commercial departures in April 2018.

They never came. At the end of 2016 the opening was postponed, because the airspace and approach routes the airport needed were not ready. Then, in May 2019, the Council of State ruled on the country's nitrogen permitting system, and the ground shifted under projects across the Netherlands. Lelystad Airport now needed a nature permit for the operations it had been built to host. For years, the low-altitude routes on which its flights would depend remained contested as well. By 2026 the route problem had largely been resolved; the disputed low-level segment had been removed and the new routes allowed continuous climbs and descents. But three conditions still stood between the finished building and its first commercial flight: a nature permit, an amended airport decree, and completed operational preparations. In April 2026 the government set a new ambition—around 10,000 flights a year from October 2027. A month later, parliamentary papers recorded that there was still no clear prospect of an irrevocable nature permit.

It would be easy to tell this as a story of a bad decision, and this report will not. Whether Lelystad should have been built is a question of costs and benefits that hindsight cannot settle, and the constraints that stalled it changed while it was being built. What the airport illustrates is something more precise and more widespread. In Lelystad, the asset came first and the feasible future came later—so much later that, eight years after the planned opening, it still has not fully arrived.

The Netherlands is full of stranger versions of the same pattern. For roughly a decade, the country's national airport operated under a formally temporary arrangement in which the state declined to enforce the legally binding noise limits, while it waited for a replacement regime to be written into law. Hundreds of megawatts of subsidised solar and wind projects were approved before anyone had checked whether the grid could carry their electricity, and found that it could not. And for four years, nitrogen permits for housing, roads and farms were granted partly on the strength of ecological improvements that had not yet happened, until the country's highest administrative court ruled that such future improvements could not be spent in advance.

These are not the failures of a country that ignores constraints. They are the failures of a country so accomplished at engineering its way through constraints that it has sometimes committed to a future before securing it. That is what this report calls **feasibility debt**.

1.2 The Pressure–Commit–Bind–Repay Loop

Each report in this series identifies a characteristic motion. Japan's architecture preserves an inherited equilibrium; Korea's competes; Bhutan's races to keep its accelerating institutions connected to its objective. The Dutch motion is a cycle of borrowing and repayment, and it has four stages.

Pressure. The Netherlands is one of the most densely populated and intensively used countries in the world, and the demands on it are simultaneous. It needs around 100,000 new homes a year. It is electrifying its industry, its transport and its heating. Its national airport is a hub of European importance. Its agriculture is among the most productive on earth. Each of these demands is legitimate and urgent, and each presses against a physical, ecological and legal environment that is already densely occupied.

Commit. Under pressure, commitments are made: subsidies awarded, permits granted, runways built, operations continued. Some of these commitments rest on capacity that exists and is secured. Others rest on capacity that is expected—ecological improvement that measures are expected to deliver, grid capacity that operators are expected to build, a legal regime that ministers are expected to enact. The commitment is real and immediate. The capacity it depends on is assumed.

Bind. Eventually the real boundary asserts itself, and the mechanism that enforces it varies. For nitrogen, it was a court: the Council of State ruled in 2019 that future ecological gains could not be credited in advance. For the grid, it was physics: the cables could not carry what had been connected to them, and projects that had been awarded subsidies could not be connected. For Schiphol, it was a combination of courts, European law and the patience of residents living with noise beyond the legal limits. The boundary was always there. What changed was that someone, or something, enforced it.

Repay. When the boundary binds, the debt comes due, and it is paid in delay, redesign, compensation, restriction or cancellation. Projects wait in queues. Permits are revoked or cannot be issued. Farms are bought out. Flights are reduced. And the conflict over who bears the cost of repayment feeds back into the next round of pressure—often in the form of demands to commit again, faster, on capacity that is once more assumed rather than secured.

The loop is not inevitable. The Netherlands contains, in its water management, a demonstrably different way of governing under uncertainty, and in several domains the country is already learning to test feasibility before it commits. But where the loop has operated, it has proved costly, slow to unwind and bitterly contested.

1.3 Feasibility Debt Defined

Feasibility debt accumulates when durable commitments are made before the ecological, infrastructural, fiscal, administrative or legal capacity needed to execute them has been secured across the relevant implementation horizon. Its defining feature is simple: **commitment precedes durable feasibility**.

The word *durable* matters. A project can be feasible on the day it is approved and infeasible by the day it is built, because other projects consume the same capacity, because the law changes, or because the ecological assumptions on which it rested turn out to be wrong. Feasibility is not a moment; it is a path. A commitment is only safe if a feasible path runs all the way from the decision to the operation it authorises.

The Dutch evidence shows three distinct forms of the debt.

Borrowed future capacity. Capacity expected to be created in the future is treated as available today. The Programmatic Approach to Nitrogen (PAS), in force from 2015 to 2019, allowed nitrogen-emitting activities to be authorised partly on the basis of ecological improvements expected from future measures. Under European nature law, which requires certainty that a protected habitat will not be harmed, that expectation was not enough. When the Council of State ruled in 2019, the borrowed capacity disappeared, and with it the permission basis for a large number of projects.

Unsecured implementation capacity. A commitment is made before the capacity needed to carry it out is secured through the period required to execute it. Before the autumn of 2019, the national subsidy scheme for renewable electricity could award support to solar and wind projects without any project-specific check that the grid could carry their output. By December 2019, Parliament recorded around 700 megawatts of subsidised projects that could not obtain a connection or transport capacity. Lelystad Airport, whose physical infrastructure was completed before its airspace, its nature permit and its operating decree were secured, is a bounded example of the same form.

Anticipated legalisation. Current operations rely on a legal regime that is expected but not yet in force. From 2015, Schiphol operated under a new system of runway use designed to reduce overall noise nuisance, which the government intended to write into the legally binding airport decree. Until that happened, the aviation inspectorate was instructed not to enforce exceedances of the existing legal noise limits that resulted from following the new system. The arrangement was meant to be a bridge. As of September 2026, it has not been fully retired.

These three forms look different, but they share a structure. In each, the country acted on the future as if it had already arrived. And in each, when the future failed to arrive on time, someone had to pay.

It matters what this definition does not claim. It does not claim that feasibility debt explains most of what goes wrong in Dutch governance; the evidence establishes that it recurs across very different domains, not that it dominates them. It does not claim that the decisions involved were irrational; several were reasonable

bets under genuine pressure. And it does not claim that the courts caused the crises in which the debts came due. The ecological, physical and legal boundaries existed before any judge enforced them. The courts called in debts that had already been incurred.

1.4 A Country Built Against Constraints

To understand why feasibility debt is a Dutch paradox rather than a Dutch habit, it is necessary to understand what the Netherlands has achieved in the domain where constraints are hardest of all.

Nearly 60 percent of the country could be exposed to flooding. On the night of 31 January 1953, a storm surge breached the dikes of the south-west and killed more than 1,800 people. The response defined the modern Dutch state's relationship with its environment. The Delta Works that followed were among the largest engineering projects of the twentieth century. And over the following decades, Dutch water management evolved from building defences against a known threat into something more sophisticated: a system for governing a threat whose future scale cannot be known.

The Delta Act, the Delta Programme, the independent Delta Commissioner and the dedicated Delta Fund form an architecture built on a premise that the rest of this report will return to repeatedly: **the future is uncertain, so preserve more than one path to it.** Dutch water planners developed an approach now used around the world—adaptive delta management, based on what are known as dynamic adaptive policy pathways. It identifies *tipping points*, the conditions under which a current strategy will stop working. It sets decision points *before* those tipping points, allowing for the years it takes to build an alternative. And it keeps alternative pathways open, so that the country can switch course without first falling into a state from which no good option remains.

In the language of control engineering, this is a method for preserving what is called **recursive feasibility**: the property that every decision leaves at least one feasible path open into the future. It is exactly what feasibility debt violates.

The same competence extends beyond flood defence. During droughts, the Netherlands allocates scarce freshwater through an established priority sequence, known in advance to every user, that protects dikes, drinking water and irreplaceable nature before agriculture, industry and shipping. When grid congestion became acute, the energy regulator created a framework for prioritising connections according to public value rather than first-come, first-served. Even nitrogen, the most contested constraint of all, has an explicit registration and allocation system for freed-up deposition space. The Netherlands does not lack the capacity to govern scarcity. It has built some of the most sophisticated scarcity-allocation systems of any state.

This is why "the Netherlands is simply too full" is the wrong diagnosis. Space is scarce, but scarce space is what the Dutch have always governed. The question is not whether the country can live inside hard constraints. It demonstrably can. The question is why, in several nationally important domains, it has

committed to futures before securing them.

1.5 Comparisons Within the Series

The Netherlands sits closest in this series to two reports that also concern timing rather than outright failure.

The report on **Bhutan** described an *Interface Race*: a country with an exceptionally clear account of what it values, building fast new economic institutions faster than it can connect them to that account. The Netherlands presents a related but distinct problem. In Bhutan, the risk is that interfaces lag behind actuators. In the Netherlands, commitments have outrun feasibility. Both are problems of synchronisation between a slow process and a fast one. But where Bhutan's challenge lies in keeping new institutions answerable to a known objective, the Netherlands' lies in keeping ambitious commitments inside a feasible future.

The report on **Finland** described a *Throughput Constraint*: a highly competent system that could not transform itself as quickly as its circumstances demanded. The Netherlands shares Finland's competence and some of its frustrations—procedures that take years, permitting systems that feel immovable. But the Dutch evidence suggests that speed is not the core problem. Faster procedures would not have created ecological capacity that did not exist, grid capacity that had not been built, or a legal regime that had not been enacted. In several of the cases examined here, the difficulty was not that the Netherlands moved too slowly, but that it committed too early.

The report on **South Korea** offers a more distant contrast. Korea's difficulty lay in its objective: a society whose many values had collapsed, in practice, onto a few rankings. The Netherlands does not suffer from a narrow objective. It suffers from a mismatch between the tempo of its commitments and the tempo at which the capacity to honour them can be secured.

1.6 The Netherlands' Genuine Strengths

Any account of Dutch governance must begin from what the country has built. It has made land from sea and kept it dry for centuries. It maintains one of the most productive agricultural sectors in the world on a territory smaller than many countries' largest regions. Its ports, logistics networks and national airport are central nodes of the European economy. Its institutions are stable, its courts independent, its administration capable and largely incorruptible, and its political culture committed to negotiation and compromise.

It also learns. After the grid shock, the renewable subsidy scheme began requiring a transport indication from the network operator before any award, moving grid feasibility ahead of the commitment. The energy regulator introduced rules allowing unused reserved grid capacity to be reclaimed—"use it on time or lose it"—so that protecting future feasibility does not needlessly block present projects. National energy-

infrastructure programming now plans major capacity before investment decisions rather than after them. And in September 2026, the government selected five new large-scale housing locations only after testing whether nitrogen, grid capacity, drinking water and accessibility would not form major obstacles—a direct reversal of the logic that produced feasibility debt. These changes are too recent to have proved themselves in outcomes. But they show a system already moving from "commit, then solve" toward "test, then commit."

The Netherlands also holds, in adaptive delta management, the most important asset any country could have for addressing the problem this report describes: a domestic, proven, internationally respected method for governing under deep uncertainty. The transition this report proposes does not require the Netherlands to invent anything. It requires the country to apply more widely what it has already invented.

1.7 The Real Question

This report therefore does not ask whether the Netherlands can govern hard constraints. Its water management answers that question. Nor does it ask whether the Netherlands has too many rules, too little space or courts that interfere too much. The evidence supports none of those diagnoses.

It asks a narrower and more interesting question:

> ***Why does the country that pioneered adaptive planning for an uncertain sea sometimes commit on land before the future is secured—and what would it take to extend its water wisdom to the other domains that now need it?***

Part of the answer, as the following sections show, is institutional. Water governance developed a complete package of continuity, independent integration, dedicated finance, alternative pathways and early decision points that other domains have acquired only in pieces. Part of it lies in the nature of the constraints themselves. Some debts can be repaid by waiting for a cable to be laid. Others can only be repaid by permanently closing a farm near a protected habitat, and no architecture can make that loss painless. And part of it lies in politics: the rural transition that might have given nitrogen a Delta-like architecture was designed, funded and then abandoned when the losses it required proved too contested to sustain.

The transition this report describes is feasible. The Netherlands has already demonstrated, at home, the capabilities it requires. What remains unproven is whether those capabilities can be carried beyond the domain where they matured—into nitrogen and farming, into energy and housing, into aviation—where the losses are more permanent, the capacity less certain, and the politics more divided. The country that learned to make room for the river now faces the task of making room, before it commits, for the futures it promises on land.

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2. The Feasibility Debt: Structural Mechanisms

2.1 What "Feasibility" Means

Every report in this series names a capacity a country lacks or strains against. For the Netherlands, the relevant capacity is not the ability to govern constraints—Dutch water management shows that it can—but the ability to keep commitments inside a feasible future. That requires being precise about what "feasible" means.

A project in the Netherlands must be feasible in two senses at once. It must be **physically feasible**: there must be land that will not flood, water to supply it, electricity to power it, and an ecological environment that can absorb its effects. And it must be **permitted**: it must satisfy the legal requirements of nature protection, water quality, noise, environmental assessment and spatial planning, and it must survive the courts. A project can be physically possible and legally impossible, or legally permitted and physically unbuildable. Only the intersection of the two is usable.

Feasibility also has a time dimension that is easy to overlook. A project is not feasible because the conditions for it exist on the day it is approved. It is feasible if a path exists from the day of approval to the day of operation along which every condition continues to hold. Large Dutch projects take years or decades to complete, and in that time other projects consume capacity, laws change, ecological assumptions are tested and courts clarify what the law requires. This report therefore distinguishes between a feasible *moment* and a **durable feasible path**. Feasibility debt arises when commitments are made on the strength of the first without securing the second.

This framing also explains why the familiar diagnosis—the Netherlands is simply full—is too shallow. Land measured in hectares is not the operative resource. A hectare may be too flood-prone, too close to a protected habitat, too far from grid capacity, too dependent on scarce drinking water, or reserved for future dike reinforcement. What matters is land that remains usable once every relevant system has had its say. The Dutch problem is less "How much space is there?" than "How much space remains feasible after all the constraints have spoken?"

2.2 The Dense Constraint Environment

The Netherlands governs an unusually dense stack of constraints, and they are not all of the same kind. Treating them as one thing—"regulation", "red tape", "scarcity"—obscures the fact that each requires a different remedy.

Some constraints are **biophysical**. Freshwater availability is falling in dry summers as demand rises; the Delta Programme now says that optimising the existing water system will no longer suffice and that some land uses cannot remain where they are. Subsidence, salinisation of peat areas and drought on higher sandy soils increasingly impose conditions on what can happen where. No procedure can create freshwater in a drought.

Some are **ecological**, and made operational by law. Nitrogen deposition remains above what much of the country's nitrogen-sensitive protected nature can tolerate. Water quality is another: across individual criteria the Netherlands meets more than three quarters of its roughly 100,000 Water Framework Directive objectives, yet under the Directive's rule that a water body passes only if it meets every objective, no Dutch surface water body met all of them at the 2024 review, and only around 15 percent of 741 assessed bodies were in good biological condition. European law turns these ecological conditions into obligations; national permitting translates the obligations into tests; courts enforce the tests.

Some are **infrastructural**. Grid congestion now affects substantial parts of the country because demand and renewable generation have grown faster than transport capacity can be built. Drinking-water utilities estimate that around 102 million cubic metres of additional production capacity will be needed by 2030; about 19 million was added in 2025. Unlike ecological limits, these constraints can be expanded—but only with land, equipment, skilled labour, permits and years of lead time.

Some are **administrative and fiscal**: the capacity of ministries, provinces, municipalities, water authorities and contractors to plan, permit and build, and the budgets available to pay for it. And some are **jurisdictional**. Dutch physical governance is shared across central government, provinces, municipalities and water authorities, overlaid by European law, independent regulators, network operators and the courts. No single actor controls the whole feasible set.

Housing is not a constraint in this list. It is a *demand*—around 100,000 homes a year—pressing against all of them at once. That makes it an unusually good probe of the system. When a housing project stalls, the useful question is not how long it was delayed but which constraint bound first.

2.3 Three Forms of Feasibility Debt

The Dutch evidence documents feasibility debt in three independent domains—ecological, infrastructural and legal—each in a different form.

Borrowed future capacity: the nitrogen permitting system. The Programmatic Approach to Nitrogen was designed to break a deadlock. Nitrogen deposition on protected habitats was too high to permit new emitting activity without assessment, but economic development could not simply stop. PAS combined source reduction, nature restoration and a permitting framework in a single programme, and it allowed new activities to be authorised partly on the strength of the ecological gains its future measures were expected to

deliver. European nature law, however, sets a demanding standard. Under the Habitats Directive, as the EU Court of Justice has interpreted it, an authority may approve a project only when it is convinced that the integrity of a protected site will not be adversely affected, with no reasonable scientific doubt remaining. Expected future gains could not meet that standard. In 2019 the Council of State ruled that PAS could not serve as a permitting basis. The capacity that had been borrowed from the future ceased to exist in law, and the permission basis for a wide range of activities—farms, roads, housing, industry—fell with it. Among those left most exposed were the so-called PAS reporters, activities that had relied in good faith on a notification scheme under the programme and found themselves without a valid permit through no fault of their own.

Unsecured implementation capacity: renewable subsidies and the grid. Until the autumn of 2019, the national subsidy scheme for renewable electricity could award support to a solar or wind project without any project-specific check that the local grid could transport its electricity. The subsidy decision and the network decision belonged to different institutions and different planning cycles. By December 2019, Parliament recorded around 700 megawatts of subsidised projects that could not obtain a connection or transport capacity and risked missing their realisation deadlines. The government's response was to insert the missing check: from the autumn 2019 round, applicants needed a transport indication from the network operator before they could receive an award. By the spring of 2020, operators had issued more than 9,000 such indications. The correction worked, but only partly. In 2026 the government acknowledged that many solar projects that had received a transport indication were nevertheless later unable to connect, because the indication showed capacity at the moment of application but did not reserve it through the years of development. Feasibility checked at one moment had decayed before the project could use it.

Lelystad Airport, described in Section 1, is a bounded example of the same form: substantial physical infrastructure completed before the airspace, the nature permit and the operating decree it required had been secured. The evidence establishes the sequence, not the merits of the original decision.

Anticipated legalisation: Schiphol. In 2015 the government began applying a new system of runway use at Schiphol, the New Norms and Enforcement System, intended to reduce overall noise nuisance by directing flights to preferred runways. The system was meant to be incorporated into the legally binding airport decree. Until then, the aviation inspectorate was instructed not to sanction exceedances of the existing legal noise limits when those exceedances resulted from following the new system. The Council of State later described the arrangement, in legal terms, as the conditional toleration of non-compliance with the statutory operating conditions in anticipation of their future legal embedding.

What makes the case so revealing is its duration. In 2021 the inspectorate warned the minister that the arrangement had become legally vulnerable and that timely legalisation was not in prospect. In 2022 the government decided in principle to end it. Exiting then became entangled in European aviation law: in July 2024 the Supreme Court ruled that measures reducing airport operating capacity must follow the EU's "balanced approach" procedure, and in March 2024 the Hague District Court found that the state had failed to protect residents adequately and must restore effective legal protection. A partial revision of the airport

decree in 2025 introduced annual limits of 478,000 flights and 27,000 night flights, but did not end the anticipatory arrangement. In September 2026 the Council of State published its advice on the new decree intended finally to do so. A bridge built in 2015 was still standing more than a decade later.

The Schiphol case should not be generalised beyond itself. The evidence does not show that anticipatory enforcement is a common instrument of Dutch administration. What it shows is that, in a nationally important domain, the state operated for roughly a decade on legal feasibility it expected to acquire rather than feasibility it had—and that the longer the arrangement lasted, the more airlines, schedules and business models were built around the borrowed space, making it harder to return.

2.4 How the Debt Is Called In

Feasibility debt does not announce itself. It becomes visible only when something enforces the boundary that the commitment had assumed away. The Dutch evidence shows several different enforcers.

The most visible are **the courts**. The PAS ruling of 2019 removed a permitting architecture built on uncertain future gains. In December 2024 the Council of State changed the treatment of internal offsetting—netting a project's new emissions against existing ones—with immediate effect on pending and future permit procedures. In January 2025 the Hague District Court ordered the state to meet its own statutory 2030 nitrogen objective, that half of nitrogen-sensitive protected habitat be brought below its critical threshold, giving priority to the most overloaded areas and attaching a penalty of €10 million if the target is missed; the judgment remains executable while the state's appeal proceeds. Beyond nitrogen, the Supreme Court's 2019 *Urgenda* judgment upheld an order requiring the state to reduce greenhouse-gas emissions by at least 25 percent below 1990 levels by the end of 2020.

It is tempting to read this record as courts obstructing government. The evidence supports a different reading. The courts did not create nitrogen deposition, climate change or noise. They enforced commitments the state had itself made in law, and they refused to accept uncertain future mitigation as a substitute for present compliance. In control-engineering terms, they acted as **intertemporal constraint enforcers**: the component of the system that stops the present from spending capacity that belongs to the future. Whether one welcomes or resents that role, it is a function, not a malfunction.

But feasibility debt does not require a court. On the grid, the enforcer was **physics**: cables and substations that could not carry what had been committed to them, and network operators obliged to say so. In infrastructure programmes, the enforcer is often **finance and delivery capacity**. In 2023 the Ministry of Infrastructure and Water Management paused seventeen road and waterway projects in its national infrastructure programme because of the accumulated pressure of budget shortfalls, nitrogen permitting

problems and shortages of staff and contractors; five more were paused in 2025, chiefly because of nitrogen. By 2026 the ministry expected to restart roughly one paused project a year, as money, permit space and delivery capacity allowed. The first, the A27 between Zeewolde and Eemnes, restarted in January 2026.

The enforcers differ, but the structure is the same. A commitment is made against capacity that is assumed. The boundary binds. The debt comes due.

2.5 Why Some Debts Are Harder to Repay

Feasibility debt explains how the system gets into trouble. It does not explain why some debts are repaid through a queue and others through a political crisis. Three properties of the underlying constraint account for much of the difference.

Certainty. How reliably can the available capacity be known? Grid transport capacity is measurable. The water available in a given drought can be observed. Nitrogen "permission space" is different in kind: it depends on where emissions fall, which habitat receives them, how that habitat responds, whether a reduction is permanent, whether it is additional to what nature recovery already requires, and how the courts interpret each of these questions. The quantity being allocated is itself uncertain—and under a legal standard that demands certainty, uncertainty cannot be spent.

Permanence. Does adjustment mean waiting, or losing something for good? A delayed grid connection is costly but temporary; the project eventually connects. A summer irrigation ban expires in autumn. Closing a livestock farm near a protected habitat, by contrast, is permanent: it ends a business, often a family's multigenerational livelihood, and changes the character of a region.

Spatial specificity. Can capacity in one place substitute for capacity in another? Grid capacity is geographically constrained, but new cables can move the boundary. Nitrogen deposition matters only in relation to a specific protected habitat; a reduction in one province does nothing for a habitat in another. The nitrogen registration system therefore tracks deposition space separately around each protected area, and in 2026 new space was described as limited precisely because an additionality test first reserved much of the reduction for ecological recovery.

Taken together, these properties explain why the same basic error—committing before feasibility is secured—produces very different politics. **Grid debt** involves relatively certain, expandable capacity and mostly temporary losses. It is repaid through queues, priority rules, network expansion and flexible contracts. Painful, but tractable. **Nitrogen debt** involves uncertain capacity, permanent losses and extreme spatial specificity. It is repaid through farm buy-outs, habitat recovery and years of litigation, and it has convulsed Dutch politics. **Schiphol's debt** sits in between: its repayment requires reducing operations that airlines and passengers have come to depend on, under European procedures that constrain how quickly it can happen.

Water provides the clearest test within a single domain. Temporary drought allocation works: the Netherlands has an established priority sequence, known in advance, and during the 2022 drought it imposed real restrictions on agriculture and industry. But the permanent land-use changes that the Delta Programme now says are necessary are proving much harder. Implementation of the freshwater strategy has been delayed by complex procedures, opposition to land purchases, shortages of labour and the termination of the national rural-areas programme. The same mature architecture that allocates temporary loss well struggles when the loss becomes permanent. This report treats these three variables as an analytical model, not a statistical law. But they point consistently in the same direction.

2.6 Constraints at Different Scales

Does the Netherlands suffer from a "constraint stack"—projects trapped by several hard limits at once? The answer depends on the scale at which one looks.

At the level of **individual projects**, the evidence shows several patterns rather than one. Some projects are held by a single dominant constraint and proceed once it is resolved. The Porthos carbon-storage project lost its reliance on a construction-phase nitrogen exemption in November 2022; after a project-specific assessment, the Council of State concluded in August 2023 that harm to the eight relevant protected areas could be ruled out, and the project went ahead. The ViA15 road project spent seven years in litigation; once other objections had been resolved, nitrogen remained the decisive issue until October 2024, when the project became final. Other projects face several independent constraints that can be resolved separately. Utrecht's Merwede district, 6,000 homes, faced a grid that could not supply conventional electrification; it responded with a collective thermal-storage and energy-sharing system through which, according to government documentation, 4,250 homes plus neighbourhood facilities place less demand on the grid than about 250 conventionally electrified homes would. A separate legal appeal against the spatial plan delayed the project by about two and a half years. Construction began in March 2025.

At the level of **long-lived megaprojects**, the constraints move. Lelystad's binding condition was airspace, then nitrogen, then a conjunction of nitrogen, regulation and operational readiness. The composition of the stack changed as the years passed—a direct consequence of committing to a project whose implementation horizon was longer than the stability of the conditions it depended on.

At the level of **programmes**, genuine stacking is clear. The seventeen paused infrastructure projects were not all blocked by every constraint at once; some were more constrained by nitrogen, others by money or staff. But taken together, the programme contained more desired projects than could fit inside the combined envelope of finance, permit space and delivery capacity available. The whole portfolio was infeasible even though no single constraint explained it.

A national study of urbanisation and energy published in 2024 identified a subtler form of propagation. Transforming an urban area into housing can require an existing business to move to a new site. If grid congestion delays the new site, the business cannot move, and the housing cannot be built. A constraint in one place blocks progress in another, through a chain of dependencies that no single project plan captures. The same study found that energy had long been treated at the strategic level as a problem that would ultimately be made solvable, while water and soil were regarded as genuine location constraints. That assumption—that feasibility can be completed after the decision—is precisely the one feasibility debt punishes.

2.7 The Legitimacy of Permanent Loss

The hardest feasibility debts require someone to give something up permanently. That makes legitimacy—whether those who bear the loss accept the process that assigns it—part of the constraint set. A technically feasible plan that cannot secure durable compliance is not operationally feasible.

Nitrogen shows this most clearly. Agriculture is a major source of nitrogen deposition, and many high-emitting farms lie close to sensitive protected habitats. In 2022 the government launched an approach to the highest emitters based on voluntary choices to stop, relocate or dramatically reduce emissions, with the resulting reductions intended partly to restore nature and legalise PAS reporters. It proposed a National Programme for Rural Areas, implemented through provincial area programmes, and reserved around €24.3 billion, partly through a dedicated transition fund.

The politics that followed were among the most intense in recent Dutch history. Large farmer demonstrations opposed the nitrogen plans. The Farmer–Citizen Movement (BBB), which emerged from that environment, won the March 2023 provincial elections. In June 2023, attempts to reach a broad agricultural agreement collapsed when the largest farmers' organisation withdrew. The following government ended the National Programme for Rural Areas in 2024, and in November of that year the legislation for the transition fund was withdrawn, with the €20.5 billion still associated with the programme to be deployed differently. Several provincial programmes were stopped or reconsidered. The underlying obligations—nitrogen, water quality, nature, climate—remained.

It would be wrong to read this as farmers simply refusing change. When compensated exit was offered on acceptable terms, demand could be strong: a cessation scheme for smaller livestock sectors in 2024–25 was oversubscribed within a day, and the government added €78 million so that every qualifying applicant could take part. The problem is not that permanent loss can never be negotiated. It is that permanent, concentrated losses—of livelihoods, property, family continuity and regional identity—require an allocation process that those affected regard as fair, adequately compensated and stable enough to plan around. The rural transition was an attempt to build that process. It was designed, funded and then abandoned. The debt remained.

Paper XIII of the *Governance as Engineering* series treats legitimacy as a gain parameter: the degree to which the governed comply and cooperate multiplies the effectiveness of every institutional choice. The Dutch rural transition is a case in which that parameter fell low enough that a technically coherent programme could not be sustained.

2.8 The Cultural Operating System

Every report in this series examines the cultural operating system beneath the institutional hardware. Three features of Dutch political culture bear directly on feasibility debt.

The first is ***maakbaarheid***—makeability, the conviction that society and its environment can be deliberately shaped. It has deep roots in a country that manufactured much of its own territory; the Dutch saying that God created the world but the Dutch created the Netherlands expresses it exactly. *Maakbaarheid* is the source of the Delta Works and of Dutch excellence in engineering, planning and logistics. It is also, in its less disciplined form, the source of the assumption that feasibility can be made later: that the grid will be built, the ecology will recover, the law will be written. The confidence that makes Dutch ambition possible also makes it easy to commit before the future is secured.

The second is the **polder model**, the Dutch tradition of consensus-building through negotiation among government, employers, unions and civil society, named after the cooperation once required to keep reclaimed land dry. Consensus is a real strength: it produces durable settlements, broad ownership and stable institutions. But it has a characteristic weakness in the face of permanent, concentrated losses. When no settlement can be reached that everyone accepts, consensus culture tends to postpone the decision rather than impose it. Postponement is exactly what feasibility debt makes possible: an assumed future capacity allows more parties to be promised something now, and the allocation of loss to be deferred.

The third is ***gedogen***, the long-standing Dutch practice of formally tolerating what is formally prohibited, of which the country's cannabis policy is the most familiar example. The evidence in this report does not show that anticipatory enforcement at Schiphol flowed from *gedogen*, and it should not be read as doing so. But the resonance is worth noting. In a political culture comfortable with a managed gap between the letter of the law and its enforcement, a decade of tolerated non-compliance while awaiting a new legal regime may have seemed less exceptional than it would elsewhere.

None of these cultural features is a flaw. Each is a strength that, in the specific conditions of dense constraints and long horizons, can tilt the system toward committing first and securing feasibility later.

2.9 Why Water Is Different—and How the Mechanisms Fit Together

The central paradox of this report is that the Netherlands governs its hardest constraint, water, through an architecture explicitly designed to avoid feasibility debt, while accumulating such debt elsewhere. A systematic comparison of five domains—water, nitrogen, the grid, aviation and housing—explains much of the difference, though not all of it.

It is not that water is monitored and the others are not. Nitrogen has a statutory programme running to 2035, with annual reporting on deposition, two-yearly reporting on measures and six-yearly reporting on habitat condition. Grid congestion has a national action programme with regular progress reports. Housing now has statutory programmes at several levels of government and a national construction monitor updated every six months. Aviation has annual enforcement reporting. Monitoring alone does not distinguish water.

What distinguishes water is the **complete package**. The Delta architecture combines, at the same time: statutory continuity through the Delta Act; an annual decision cycle; an independent office, the Delta Commissioner, responsible for coherence across governments and time horizons; a dedicated long-horizon fund; explicit treatment of uncertainty as unavoidable; alternative pathways maintained in advance; signals and reviews capable of changing the chosen path; and decision points set early enough to allow for implementation lead time. The other domains have acquired pieces of this package—targets, monitoring, programmes, increasingly early feasibility checks—but none has the whole. Nitrogen came closest to acquiring it: the rural transition was, structurally, an attempt to build a Delta-like architecture for the countryside, with a long horizon, area programmes and a dedicated fund. That attempt was abandoned.

But architecture is only half the explanation. Water governance has been most successful where options could be kept flexible, measures staged and collective benefits were large. Its own difficulties are growing precisely as it moves toward permanent land-use change and toward a financing gap: the 2027 Delta Programme reports that around €28 billion is available against roughly €40 billion of required measures through 2050, a shortfall of about €12 billion, and says that technical interventions alone will no longer suffice. The Delta model is strongest where option preservation is technically possible. Its hardest test begins where adaptation requires permanent redistribution—which is where nitrogen has been all along.

The mechanisms therefore fit together into a single account. A **dense constraint environment**, **long implementation horizons** and **pressure to commit** lead, in domains without a complete option-preserving architecture, to commitments that precede durable feasibility. That is **feasibility debt**. The debt is called in by courts, by physics, by finance or by ecology. How painfully it must be repaid depends on the **certainty, permanence and spatial specificity** of the constraint. Where repayment requires permanent, concentrated losses, **legitimacy** becomes part of the constraint set, and the institutions needed to allocate those losses must themselves survive political conflict. The cultural strengths of *maakbaarheid* and the polder model make it easy to commit first and settle losses later.

And the counter-model is already Dutch. **Recursive feasibility**—preserving at least one credible feasible path through the full implementation horizon—is what adaptive delta management does for water. The question for the rest of this report is how that capability can be carried, by function rather than by imitation, into the domains that now need it.

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3. What Building Recursive Feasibility Would Look Like

3.1 The Principle: Preserve a Feasible Path, Not Every Option

The obvious response to feasibility debt is to demand certainty before any commitment. It would be the wrong response. A country that waited until every constraint was secured before committing to anything would build nothing, and the Netherlands cannot afford to stop building. It needs homes, grid capacity, energy infrastructure and the transformation of its countryside, and it needs them on timescales shorter than the time it takes to eliminate every uncertainty.

The opposite response—reserving capacity generously for every commitment, so that nothing can later be displaced—fails in a different way. Dutch grid regulation has already discovered that capacity reserved but unused is capacity denied to someone else. Over-protection of future feasibility creates artificial scarcity in the present.

The principle that follows is the one Dutch water management already applies:

> ***Do not commit to a future before securing at least one credible feasible path to it—and keep that path open only as long as it is actually needed.***

This is **recursive feasibility**: the property that every decision leaves the system in a state from which a feasible way forward still exists. It does not require perfect prediction. Adaptive delta management works precisely because it does not pretend to know the future; it maintains several possible futures and decides in advance how it will recognise which one is arriving. Nor does it require maximum reservation. It requires *dynamic preservation of options with disciplined release*.

The sections that follow describe what that principle would mean in practice outside water. They share one constraint, stated plainly by the evidence: the Delta architecture cannot simply be copied. Nitrogen, the grid, aviation and housing differ from water in their authorities, their legal frameworks and the nature of the losses they impose. What can be transferred is not the institution but its functions.

3.2 The Feasibility-Pathway Assessment

The most concrete instrument this report proposes is a **feasibility-pathway assessment** for major, long-horizon national commitments: large infrastructure projects, national housing locations, major energy investments, significant changes to aviation regimes, and national subsidy schemes that induce long-lived private investment.

The assessment would answer nine questions before commitment:

1. **Which constraints must remain satisfied** from the moment of commitment through construction and operation?
2. **Which of the required capacity exists now**—ecological, infrastructural, fiscal, administrative and legal?
3. **Which depends on future delivery**, and by whom?
4. **How certain is that delivery**, and on what evidence?
5. **What capacity or options must be conditionally preserved** so that the chosen path remains executable?
6. **When must feasibility be revalidated**, and against which conditions?
7. **What alternative pathway remains** if the preferred one closes?
8. **When should reserved resources be released** if the commitment does not progress?
9. **Who bears permanent losses** if the debt must be repaid, and how?

None of these questions is new to Dutch planners. Many are already asked in environmental assessments, infrastructure planning and the new housing-site screening. What is new is asking all of them, together, before commitment, and recording the answers in a form that can be checked later. The ninth question matters most. The hardest feasibility debts have been those whose repayment required permanent losses that no one had agreed to bear. An assessment that forces the question to be answered before the commitment is made would not make those losses disappear, but it would stop them from being deferred by default.

The assessment would not decide whether a project should proceed. That remains a political choice. It would make the terms of the choice visible: how much of the proposed future is secured, how much is borrowed, and who will pay if the borrowing fails.

3.3 The Four-Rung Ladder

The assessment's questions correspond to four increasing levels of feasibility protection. Each has a Dutch precedent.

Observe. Establish what capacity exists at the moment of decision. The transport indication introduced for renewable subsidies in 2019 is the model: before an award, the network operator confirms that capacity is visible at the proposed location.

Forecast. Estimate what capacity will exist during implementation, given other claims on it. The weakness of the transport indication, acknowledged in 2026, is that it observes without forecasting: capacity visible today may be consumed by other projects before this one is built.

Conditionally preserve. Protect enough capacity, or enough alternative options, that the chosen path remains executable through its implementation window. The Delta Fund is a partial example in finance; spatial reservations for future dike reinforcement are an example in land.

Revalidate and release. Test periodically whether reserved capacity is still needed, and release it if the commitment has stalled or conditions have changed. The energy regulator's "use it on time or lose it" rules, which allow network operators to reclaim contracted transport capacity that goes unused, are the Dutch precedent. They exist because the third rung, applied without the fourth, produces its own failure: capacity hoarded against futures that never arrive, while feasible projects wait.

The ladder makes explicit a trade-off that feasibility debt conceals. Too little preservation, and commitments outrun the capacity to honour them. Too much, and capacity sits idle while others are excluded. The right level lies between, and it is found not by rule but by revalidation.

3.4 Decision Points Before Boundaries

The single most transferable element of adaptive delta management is not an institution but a habit of thought: identifying, in advance, the conditions under which a current strategy will stop working, and placing the decision to change course early enough to act before those conditions arrive.

Dutch water planners call these conditions *adaptation tipping points*. Their insight is that the decision point cannot coincide with the tipping point, because alternatives take time to build. If a dike will cease to provide adequate protection in twenty years, and a replacement takes fifteen years to plan and build, the decision must be taken within five.

Applied outside water, the same logic would ask, for each major commitment: at what point would this path become infeasible, how long would it take to build an alternative, and therefore when must we decide? For nitrogen, the tipping point might be the point at which a permitting approach becomes legally untenable. For the grid, the point at which regional demand will exceed planned capacity. For Schiphol, the point at which a provisional regime can no longer be defended. In each of the cases this report has examined, those tipping

points were foreseeable—the aviation inspectorate warned of Schiphol's legal vulnerability in 2021, years before the arrangement could be retired. What was missing was a decision point placed before them, with an alternative ready.

3.5 Transferring the Delta Package by Function

The Delta architecture works because it combines continuity, integration, finance, pathway switching and early decision points in a single package. Other domains need the functions, not the institutions.

Continuity. Water planning is protected from political cycles by the Delta Act, which requires an annual programme regardless of which government is in office. Domains prone to feasibility debt need a comparable guarantee that the feasibility assessment, the monitoring of reserved capacity and the tracking of outstanding debts continue across changes of government. That can be achieved through legislation that fixes the obligation rather than the policy.

Integration. The Delta Commissioner's function is to maintain coherence across governments, sectors and time horizons, and to report independently on progress. Nitrogen, grid and housing policy each have monitoring, but none has an actor whose explicit task is to see across the whole feasible set and say when commitments are outrunning it. That function could be assigned to an existing independent body with the right mandate rather than to a new commissioner for every field.

Finance. The Delta Fund gives water planning the ability to act on long horizons. The rural transition attempted to build an equivalent and abandoned it. Where repayment of feasibility debt requires long-term, predictable funding—above all for compensated transitions in agriculture—a dedicated, multi-year financing arrangement is a functional requirement, not a luxury.

Pathway switching. Water planning maintains alternative strategies and knows in advance what would trigger a switch. Other domains tend to commit to a single preferred path and discover its failure only when the boundary binds.

The evidence does not support creating a Delta Commissioner for every field, and it does not suggest that the institutions of water governance would succeed if transplanted wholesale. It does support making these four functions explicit wherever long-horizon commitments are made under dense constraints.

3.6 A Legitimacy Architecture for Permanent Loss

Feasibility debt is hardest to repay where repayment requires permanent, concentrated losses. No assessment, ladder or decision point makes such losses painless. What can be designed is the process by which they are allocated—so that those who bear them regard it as fair, adequately compensated and stable enough to plan around.

The Netherlands has a domestic precedent. **Room for the River**, the national programme carried out between roughly 2006 and 2015, gave land back to the country's great rivers so that they could carry higher flows safely. It lowered floodplains, relocated dikes and created bypasses at more than thirty locations. In several places it required permanent change to land use, including the relocation of farms and, in some cases, homes. It is widely regarded, in the Netherlands and abroad, as a success. It showed that Dutch governance can allocate permanent, place-specific losses when certain conditions hold: a clear public purpose, early and genuine involvement of affected communities, generous and predictable compensation, local design freedom within a national framework, and a programme stable enough to outlast the political cycle.

Those conditions are exactly what the rural transition lacked by the time it was abandoned. A legitimacy architecture for permanent loss would therefore include:

- **Predictable, adequate compensation**, backed by long-horizon finance that does not change with each coalition agreement;
- **Credible alternatives**—relocation, extensification, new income sources—so that exit is not the only option offered;
- **Transition time** matched to the life of the businesses and families affected;
- **Area-based design**, so that the form of adjustment reflects local conditions rather than a single national template;
- **Procedural stability**: rules that, once set, are not reopened by every change of government.

The oversubscribed cessation scheme of 2024–25 shows that where these conditions are met, negotiated exit can be attractive. The failure of the broader rural transition shows what happens when they are not.

3.7 Courts as Backstop, Not Sole Enforcer

In the Dutch cases examined here, courts often became the mechanism through which feasibility debt was called in. That is not a failure of the courts. It is a sign that other mechanisms did not act first.

A system in which the boundary is enforced only by litigation is a system that discovers its infeasibility late, expensively and all at once. The PAS ruling did not merely invalidate a permitting scheme; it removed the permission basis for a wide range of activities in a single judgment. Recursive feasibility aims to move enforcement earlier: to have the feasibility-pathway assessment, the revalidation of reserved capacity and the tracking of outstanding debts detect when commitments are outrunning capacity, so that adjustment happens gradually and by design rather than suddenly and by judgment.

This does not reduce the role of the courts. Independent judicial review of whether the state honours its own legal commitments remains essential, and the courts' willingness to refuse uncertain future mitigation as a substitute for present compliance is itself a safeguard of recursive feasibility. The aim is simply that the courts should be the backstop, not the first line.

3.8 Cultural Software Update: From Making Land to Making Room

Every transition in this series ultimately requires the cultural operating system to evolve without losing its strength. The Dutch operating system is unusually well suited to the task, because its central strength—the confidence that the country can shape its own environment—is exactly what recursive feasibility needs, once it is disciplined.

The evolution this report proposes is captured in the name of the programme that already embodies it. For centuries, *maakbaarheid* meant making land: pushing back the sea, draining the lakes, reclaiming the polders, and assuming that engineering would find a way. Room for the River represented a different idea. Instead of forcing the river into a channel that could no longer contain it, the Netherlands gave it space. It accepted a boundary, planned around it, and allocated the loss that followed.

That is the cultural shift feasibility debt requires: from *making* feasibility after the commitment to *making room* for it before. It keeps the Dutch confidence in design, but applies it to the design of pathways rather than the assumption of outcomes. It keeps the polder tradition of negotiation, but asks it to settle who bears permanent losses before those losses are forced, rather than after. And it keeps the Dutch capacity for pragmatic tolerance, but limits it to what is genuinely temporary, with a date by which the gap between the rule and the practice will be closed.

The Netherlands does not need to become less ambitious. It needs its ambition to be recursively feasible—to leave, with every commitment it makes, a path by which that commitment can actually be kept.

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4. The Political Immune System: What Keeps Producing Debt

4.1 An Immune System Without Villains

Every report in this series examines the forces that protect an existing pattern from change. In the Netherlands, no one defends feasibility debt. No minister argues that permits should rest on ecological gains that may never arrive, or that subsidies should be awarded for projects the grid cannot carry. The debt is not a policy anyone chose. It is a by-product of pressures that each have their own legitimacy.

That makes the Dutch immune system unusual. It does not protect a dysfunction that benefits a powerful group. It protects a *habit*: the habit of resolving a present conflict by assuming a future that will make it disappear. The forces described below sustain that habit. None of them requires bad faith, and several of them are among the strengths of Dutch political life.

A caution applies throughout. This report does not claim that Dutch policymakers deliberately create feasibility debt in order to avoid hard choices. The evidence does not support claims about motive. What it supports is a structural observation: feasibility debt *can* allow difficult allocation decisions to be postponed, and the forces described here make that postponement easier to accept.

4.2 The Pressure to Promise

The first force is the sheer weight of simultaneous demand. The Netherlands needs around 100,000 new homes a year. It has committed to rapid decarbonisation, which requires electrifying industry, transport and heating and building vast new energy infrastructure. Its national airport is a hub on which much of its economy depends. Its agricultural sector is among the most productive in the world. Each of these demands has a constituency, a target and a political deadline.

Under that pressure, the cost of saying "not yet" is immediate and visible: a housing site not designated, a renewable project not subsidised, a flight schedule curtailed. The cost of committing on assumed capacity is deferred and diffuse: a permit that may be challenged later, a connection that may not be available when the project is built, a legal regime that may not arrive in time. Any political system facing that asymmetry will tend to prefer the commitment.

The asymmetry is sharpened by the length of Dutch implementation horizons. A political term lasts four years, and often less. A housing location, a grid reinforcement or an airport regime takes a decade or more to complete. The benefits of committing accrue within the term; the costs of the debt fall due in the next one, or

the one after.

4.3 Uncertainty That Looks Like Slack

The second force is subtler, and it lies at the heart of the mechanism. When capacity is known to be zero, a government must confront scarcity directly. When capacity is known to exist, allocation is straightforward. The dangerous case is the one in between: when capacity *might* exist later.

Uncertainty of that kind has political value. It allows more constituencies to be promised something now, on the expectation that the future will supply what the present cannot. The nitrogen permitting system relied on ecological improvements that were expected but not certain. The renewable subsidy scheme relied on grid capacity that would be built but had not been. The Schiphol arrangement relied on a legal regime that would be enacted but was not yet in force. In each case, the uncertainty about future capacity was treated, in practice, as if it were capacity.

The principle can be stated simply: **uncertainty can masquerade as slack.** It is not a failure of forecasting. It is what happens when a system under pressure is allowed to count the upper range of its expectations as present resources. The legal standard that ultimately defeated PAS—that there must be no reasonable scientific doubt—exists precisely to prevent that substitution in nature protection. Most other domains have no equivalent rule.

4.4 Consensus and Postponement

The third force is the polder model itself. Dutch political culture values consensus: settlements negotiated among government, sectors and civil society, broadly owned and durable once reached. That tradition has served the country extraordinarily well, and it remains one of its great strengths.

But consensus has a characteristic difficulty with permanent, concentrated losses. When a settlement requires some parties to give up their livelihoods permanently so that others may proceed, it is often impossible to reach one that everyone accepts. The consensus instinct is then to keep negotiating—to seek a formula that allows everyone to continue. Feasibility debt supplies that formula. An assumed future capacity allows farmers, builders, road users and nature to be promised space simultaneously, and defers the question of who will actually give way.

The attempt to reach a broad agricultural agreement in 2023 illustrates the limit. The negotiation was an exercise in consensus-building of exactly the Dutch kind. It failed when the largest farmers' organisation withdrew. The underlying question—who would bear the permanent adjustment that nitrogen and water quality required—was not settled, and the obligations that made it necessary remained.

4.5 No One Holds the Whole Feasible Set

The fourth force is structural. Dutch physical governance is distributed across four tiers—central government, provinces, municipalities and water authorities—and overlaid by European law, independent regulators, network operators, drinking-water companies and the courts. Each actor governs its own constraint competently. None is responsible for the whole.

A municipality can designate a housing site. A province can support a regional development plan. A network operator can still lack capacity. A water company can still lack supply. European nature law can still require assessment. A court can still invalidate the permit basis. Every decision in the chain may be individually rational, and the project may still be infeasible, because no one was charged with checking the intersection before the commitment was made.

This fragmentation is not a design error. Distributed authority protects values that a single optimiser might sacrifice, and it keeps decisions close to the places they affect. But it means that the feasible set is, in effect, ownerless. Water is the exception: the Delta Commissioner exists precisely to see across the tiers and across time. Elsewhere, the function that would detect feasibility debt before it accumulates has no holder.

4.6 Political Discontinuity

The fifth force is the instability of the institutions that would repay the debt. Recursive feasibility requires continuity: commitments, reservations and transition arrangements that outlast the government that made them. Recent Dutch politics has not supplied it. Coalitions have been complex and fragile, and several cabinets have ended before completing their terms.

The rural transition is the clearest casualty. One government designed a long-horizon programme with a dedicated fund of the kind that might have given nitrogen a Delta-like architecture. The next ended the programme and withdrew the fund legislation, redirecting the money it had reserved. Whatever the merits of either government's approach, the effect on feasibility was the same: provinces that had begun area programmes stopped or reconsidered them, farmers who had been told one thing were told another, and the underlying obligations continued to accumulate.

Water is again the contrast. The Delta Act was designed to insulate flood protection from exactly this kind of discontinuity, and it has done so across many changes of government. The lesson is not that one government was right and the other wrong. It is that the repayment of long-horizon feasibility debt requires institutions that survive the political cycle, and that in most domains such institutions do not yet exist.

4.7 Lock-In Around Borrowed Space

The sixth force operates once debt has been incurred. Activity organised around borrowed feasibility acquires its own constituency. The longer the debt remains outstanding, the more investments, business models, jobs and expectations depend on it, and the more costly repayment becomes.

Schiphol is the clearest example. During a decade of anticipatory enforcement, airlines built schedules, networks and investments around an operating regime that was not fully legal. Returning to a legally secure envelope now means reducing operations that many actors have come to depend on, under European procedures that require careful balancing before capacity can be cut. The arrangement that was meant to bridge a short gap became something that people built on.

The PAS reporters show the same dynamic from the other side. They were farmers and businesses that had relied in good faith on a notification scheme the state itself had created. When the scheme fell, they were left without valid permits. Their legalisation has been a priority for the allocation of freed nitrogen space ever since, competing with housing and infrastructure for a limited pool. The debt the state incurred became a burden carried by people who had trusted it.

4.8 Who Benefits from Feasibility Debt—Named Honestly

The series names the beneficiaries of each immune system. In the Netherlands the list is unusual, because almost everyone who benefits from feasibility debt when it is incurred is also harmed by it when it falls due.

Governments and coalitions benefit from being able to promise housing, energy, mobility and agricultural continuity simultaneously, and from deferring the allocation of permanent losses beyond their term. They also inherit the debts their predecessors incurred.

Sectors that operate on borrowed space—the aviation sector at Schiphol during anticipatory enforcement, activities permitted under PAS—benefit from continuing to operate while the legal or ecological basis is incomplete. They bear the cost when the space is withdrawn.

Developers and investors benefit from commitments made early, before constraints are fully tested. They are also among the first to lose when projects stall: the renewable developers left holding subsidies without connections, the builders whose sites could not be permitted.

Consumers and residents benefit from homes, electricity and flights delivered sooner. Some of them—residents living with noise beyond legal limits, communities near overloaded habitats—bear the cost of the gap between what was committed and what was lawful.

Farmers occupy both positions at once. Some benefited from permits that PAS made possible. Many now face the permanent adjustment that repaying the nitrogen debt requires.

The honest conclusion is that feasibility debt is not a transfer from losers to winners. It is a transfer from the future to the present. The same actors often sit on both sides of it, separated only by time.

4.9 The Narrative Strategy

How the transition is framed will determine whether it is heard as caution imposed on an ambitious country or as the extension of something the country already does well. Five principles follow.

Frame recursive feasibility as Dutch water wisdom applied at home. The Netherlands is internationally admired for how it governs water under uncertainty. The case for feasibility-pathway assessments is strongest when presented not as a new constraint but as the application of the Delta method to the other domains that now need it.

Frame it as protection for those who trust the state. The PAS reporters and the renewable developers left without connections were harmed by commitments the state made on their behalf. Securing feasibility before commitment is, above all, a way of keeping the state's word.

Do not frame it as slowing down. The argument against feasibility debt is not that the Netherlands should do less. It is that the country should commit in ways it can keep, so that projects proceed rather than stall halfway.

Name the permanent losses early. The hardest debts arose where no one had agreed who would bear the cost of repayment. The narrative should make clear that settling this question before commitment—through fair, predictable and well-funded transition arrangements, as Room for the River did—is kinder to those affected than leaving it to courts and crises.

Present the courts as allies of good planning. Framing the courts as obstacles invites the conclusion that the answer is to weaken them. The better frame is that courts have repeatedly enforced commitments the state made itself, and that the best way to avoid being overruled is to plan so that the law's boundaries are respected from the start.

The immune system cannot be defeated by argument alone, because it rests on real pressures and real strengths. But most of it is sustained by actors who are themselves damaged by the debts they help create. The transition does not need to overcome them. It needs to show them that a country that secures its futures before committing to them delivers more of what they want, not less.

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5. Working with the Grain: Transition Architecture for the Netherlands

5.1 The Principle: Start Where the Learning Already Exists

The Netherlands does not need to be persuaded that feasibility matters. In several domains it has already begun to test feasibility before committing, and the transition should build from those beginnings rather than announce a new doctrine.

The evidence is recent but real. Since 2019, renewable subsidies have required a transport indication from the network operator before an award. The energy regulator's "use it on time or lose it" rules allow unused grid capacity to be reclaimed. National energy-infrastructure programming now plans major capacity before investment decisions rather than after. And in September 2026 the government selected five large housing locations only after testing whether nitrogen, grid capacity, drinking water and accessibility would not form major obstacles. Each of these is a partial step up the ladder described in Section 3. None is yet complete, and none has had time to prove itself in outcomes. Together they show a system already moving from "commit, then solve" toward "test, then commit."

The transition should also respect how the Netherlands changes. Dutch reform rarely proceeds by decree. It proceeds through expertise, negotiation and incremental institutionalisation: a problem is studied by the planning agencies, debated among the parties, piloted, evaluated and then written into law once a consensus has formed. The Delta Programme itself followed that path, from the Delta Committee's advice in 2008 to the Delta Act. A transition that works with this grain will be evidentiary before it is statutory, and it will make its case through Dutch examples rather than imported models.

5.2 The Trojan Horse: Delta-Proof Decisions

Every transition architecture in this series has identified a Trojan Horse: a mechanism that carries a transformative payload in familiar packaging. In Bhutan it was GNH materiality; in Korea, fairness itself.

In the Netherlands, the Trojan Horse is the **Delta method**. Few things command such broad pride and trust in Dutch public life as the country's mastery of water. Adaptive delta management is taught around the world, exported through Dutch engineering firms and cited in international climate-adaptation policy. Presenting

feasibility-pathway assessments as *Delta-proof decisions*—the application of the country's own water wisdom to its other long-horizon commitments—carries the payload of recursive feasibility inside a frame that almost no one in Dutch politics would reject.

The framing is not a disguise. It is accurate. Tipping points, decision points set before them, alternative pathways and periodic revalidation are exactly what the Delta method does. The payload is simply the recognition that nitrogen, the grid, aviation and housing are now as constrained, and as uncertain, as the sea.

5.3 Safe-to-Fail Pilots

Four pilots would test the approach at low cost, each chosen because it is already under way or imminent.

Pilot 1: the new national housing locations. The five locations selected in 2026 have already been screened for joint feasibility at the moment of selection—the first rung of the ladder. A pilot would extend that screening across the implementation horizon: forecasting whether grid, water and nitrogen capacity will remain available as each location develops, identifying what must be conditionally preserved, setting revalidation dates, and naming the alternative if a constraint closes. If these locations subsequently show materially fewer late-stage feasibility failures than earlier developments, the case for extending the approach will make itself.

Pilot 2: major energy infrastructure. The national programme for energy infrastructure already plans capacity ahead of investment. A pilot would add the revalidate-and-release rung: explicit review dates at which capacity reserved for industrial electrification is either confirmed or released to others, avoiding both stranded commitments and hoarded capacity.

Pilot 3: the Lelystad opening decision. The government's current plan for commercial operations at Lelystad depends on three conditions that must clear together. Applying a feasibility-pathway assessment to that decision—stating which conditions are secured, which depend on future delivery, what the fallback is if the nature permit cannot be obtained, and when the decision will be revalidated—would turn the report's opening case into its first demonstration.

Pilot 4: a pre-2027 water-quality exposure assessment. The Netherlands is not on course to meet all its Water Framework Directive objectives by the 2027 deadline. A pilot would identify, before the deadline, which existing permits, plans and commitments depend on water-quality improvements that may not materialise—finding any feasibility debt *before* a court or the European Commission calls it in. This is the most time-sensitive of the four. Its value lies entirely in acting before the boundary binds.

Each pilot is inexpensive, informative whatever its outcome, and compatible with decisions the government is already making.

5.4 Scaling by Attraction

Dutch policy innovations spread through evidence and example. The Netherlands has unusually strong independent planning agencies—for environmental, economic and social assessment—whose evaluations carry weight across the political spectrum. The transition should use them.

If the pilots show that feasibility-pathway assessments reduce stalled projects, stranded investments and late legal defeats without slowing overall delivery, that evidence, evaluated independently and published, will be the most persuasive argument available. Ministries, provinces and municipalities that adopt the approach will be able to show that their commitments hold. Developers and investors, who bear much of the cost when projects stall, have every reason to prefer commitments that come with a secured path. And the courts, faced with decisions that have already tested their own legal assumptions, are less likely to become the mechanism through which feasibility is discovered.

The state's role is to measure, publish and hold up the examples that work. The Netherlands does not need to be told to plan well. It needs evidence that planning for durable feasibility pays.

5.5 Sequencing

The order of operations matters, because the transition must avoid adding to the stock of debt while it works to reduce it.

First, make the debt visible. The feasibility balance sheet proposed in Section 6 records the outstanding debts—unresolved PAS reporters, Schiphol's legal status, grid queues, paused infrastructure programmes—in one place. It changes no decision, but it makes the scale of the problem, and its movement over time, visible to Parliament and the public.

Second, stop adding to it. Apply feasibility-pathway assessments to new major commitments, beginning with the pilots. This is the cheapest and most important step: every commitment made on a secured path is a debt not incurred.

Third, build the legitimacy architecture for permanent loss. Repaying the nitrogen debt requires long-horizon, predictable arrangements for compensated transition in agriculture. Whatever form they take, they must be designed to survive changes of government, as Room for the River and the Delta Fund did.

Fourth, retire the existing debts deliberately. Schiphol's anticipatory arrangement, the legalisation of PAS reporters and the paused infrastructure portfolio should each have a published pathway to resolution, with dates and fallbacks, rather than remaining open indefinitely.

The steps overlap, and none should wait for the previous one to finish. But the direction is clear: first see the debt, then stop adding to it, then build the means to repay it, then repay it.

5.6 The Role of the Centre: Legislate the Obligation, Not the Outcome

The centre's most important contribution is continuity. Recursive feasibility requires that commitments, reservations and transition arrangements survive the government that made them, and recent Dutch politics has shown how easily they do not.

The Delta Act offers the template. It does not prescribe which dikes to build; it requires that a Delta Programme be produced every year, that a Commissioner report on it independently, and that a fund exist to pay for it. The same approach could be applied to feasibility debt. Legislation could require feasibility-pathway assessments for commitments above defined thresholds, an annual feasibility balance sheet presented to Parliament, and independent reporting on both. It would fix the obligation to assess and disclose, not the policy choices that follow.

The integration function—seeing across constraints, tiers of government and time horizons, and saying when commitments are outrunning the feasible set—should be assigned to an existing independent body rather than a new institution. The Netherlands has several candidates, from its planning agencies to its advisory councils on the living environment and infrastructure. What matters is that the mandate is explicit, the reporting public and the independence protected.

The centre should resist two temptations. The first is to create a new coordinating authority with power over every constraint. That would recreate, at national level, the problem it is meant to solve: a single optimiser overriding the distributed authorities that protect values it might neglect. The second is to treat the Delta institutions as a blueprint to be copied. Their functions transfer; their form does not.

The Netherlands built the most respected architecture in the world for governing an uncertain future. The transition proposed here asks it to do something it has done before: to recognise a new domain of deep uncertainty and to build the institutions needed to govern it before the next flood arrives rather than after.

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6. A Concrete First Step: The Feasibility Balance Sheet

6.1 The Logic of the First Step

Feasibility debt is a structural condition, and no single measure will retire it. The first step should therefore not be the most ambitious reform this report has described. It should be the one that makes every later reform easier: low in cost, unthreatening to the institutions concerned, and capable of changing how the country sees its own commitments.

Two tempting candidates fail that test. A new authority with power over every constraint would override the distributed institutions that protect values a single optimiser might neglect, and it would take years to build. An immediate legal requirement to secure feasibility before every commitment would be resisted as a brake on housing, energy and infrastructure at the moment the country most needs them.

The first step proposed here does neither. It starts from a simple observation. The Netherlands accounts meticulously for its fiscal debt. It publishes its borrowing, its interest obligations and its contingent liabilities, and it presents them to Parliament every year. It has no equivalent account of the commitments it has made against capacity it has not secured. Its feasibility debts—the permits resting on capacity that no longer exists, the operations running ahead of their legal basis, the subsidised projects without connections, the infrastructure programme that exceeds its combined envelope of money, permits and staff—are recorded, if at all, in scattered letters, rulings and progress reports.

The first step is to account for them in one place. This report calls the instrument the **Feasibility Balance Sheet**.

6.2 What the Balance Sheet Would Record

The balance sheet would list the country's outstanding feasibility debts: national commitments that currently depend on ecological, infrastructural, fiscal, administrative or legal capacity that has not been secured. For each entry it would record:

The commitment. What was promised, authorised, subsidised, built or permitted, and when.

The form of the debt. Whether it rests on borrowed future capacity, on unsecured implementation capacity, or on anticipated legalisation.

The scale. An appropriate measure of the exposure: the number of permits or projects affected, the megawatts awaiting connection, the flights operating under a provisional regime, the budget required to restart paused projects.

Who is exposed. Which actors bear the risk if the debt is called in—farmers, developers, residents, airlines, public budgets.

The likely enforcer. Whether the boundary is most likely to be enforced by a court, by physical capacity, by finance, or by ecological conditions.

The repayment profile. How certain the missing capacity is, whether repayment would require permanent losses, and how tied it is to specific places—the three properties that determine how hard the debt will be to repay.

The repayment pathway. What is being done to secure the missing capacity or retire the commitment, by whom, and by when—and what happens if that pathway fails.

Alongside the outstanding debts, the balance sheet would record **contingent debts**: commitments that are not yet in difficulty but whose feasibility depends on future delivery that is uncertain. Fiscal accounts already distinguish between debt and contingent liabilities such as guarantees; the same distinction applies here. A commitment that relies on water-quality improvements yet to materialise, or on grid reinforcement not yet built, belongs in this column before it becomes a problem.

6.3 The First Entries

On the evidence of this report, an initial balance sheet would include at least the following.

Nitrogen and the PAS reporters. Activities that relied on the notification scheme under the former nitrogen programme and were left without valid permits when it fell. Form: borrowed future capacity. Exposure: the farmers and businesses concerned. Repayment profile: uncertain capacity, frequently permanent adjustment, extreme spatial specificity.

Schiphol's anticipatory enforcement. Operations conducted since 2015 under tolerated departures from the legally binding noise limits, pending a replacement legal regime. Form: anticipated legalisation. Exposure: surrounding residents, the aviation sector and, through European procedures, the pace at which any reduction can be made.

Renewable projects awaiting connection. Subsidised projects, including those that received a transport indication but found capacity unavailable when they were ready to connect. Form: unsecured implementation capacity. Repayment profile: relatively certain and expandable capacity, largely temporary losses.

The paused infrastructure portfolio. The road and waterway projects paused since 2023 because the programme exceeded the combined envelope of finance, nitrogen permit space and delivery capacity. Form: unsecured implementation capacity at programme level.

Lelystad Airport. Completed physical infrastructure whose commercial operation still depends on a nature permit, an amended airport decree and operational readiness. Form: unsecured implementation capacity, recorded without judgement on the merits of the original investment.

Water-quality commitments (contingent). Permits, plans and land uses whose feasibility may depend on progress toward Water Framework Directive objectives the Netherlands is not on course to meet by 2027.

These entries are not an indictment. Several reflect reasonable decisions taken under genuine pressure, and some are already being repaid. Their value lies in being seen together, measured consistently and tracked over time.

6.4 The Companion Instrument: Feasibility-Pathway Assessment

The balance sheet makes existing debt visible. Its companion, the feasibility-pathway assessment described in Section 3, prevents new debt from accumulating.

The two are designed to work together. Every major new commitment above defined thresholds would undergo an assessment before it is made. Where the assessment shows that the commitment depends on capacity not yet secured, the commitment may still proceed—that remains a political choice—but it enters the balance sheet as a contingent debt, with its repayment pathway, revalidation dates and fallback recorded. If the missing capacity is secured, the entry is closed. If it is not, the entry moves from the contingent column to the outstanding one, and Parliament can see it happen.

That linkage changes the incentives without changing any decision rule. A government that chooses to commit on borrowed capacity may still do so. It simply cannot do so invisibly.

6.5 Institutional Design

The balance sheet's credibility depends on its independence and its regularity.

It should be compiled by an independent institution with standing in both public accounts and physical planning—for example, the Netherlands Court of Audit working with the national planning agency for the living environment, which already monitors nitrogen, water and spatial development. Each ministry responsible for a commitment would prepare its own entries; the independent institution would verify them, ensure consistency across domains and publish the result.

The natural moment for publication is Budget Day, when the government presents its fiscal accounts to Parliament. Presenting the feasibility balance sheet alongside the fiscal one would make a simple point: that the state's commitments against physical, ecological and legal capacity are as real as its commitments against money, and deserve the same scrutiny. The government would be required to respond to the balance sheet's principal findings, as it responds to the budget debate, without being bound by them.

The design is deliberately modest. The balance sheet creates no new authority, cancels no project and requires no immediate legal change. Its only instruments are measurement and publication. That modesty is what allows it to pass the political immune system described in Section 4. No sector loses anything when it is introduced. And once it exists, every future commitment will be made in the knowledge that its feasibility will be recorded.

6.6 Registered Predictions

This report offers a diagnosis that evidence should be able to weaken or overturn. It registers four predictions now, before their outcomes are known. The balance sheet would provide the means to track them.

P1 — Water quality after 2027. If permits, plans or land uses have relied on water-quality improvements that fail to materialise, enforcement after the 2027 deadline may expose new feasibility debt, through courts or through European infringement procedures. If lawful exemptions, phased restoration and differentiated programmes absorb the shortfall without a broad shock to permitting, the nitrogen case will look more exceptional. This report does not predict a nitrogen-style shock; it registers the test.

P2 — The new housing locations. The locations selected in 2026 underwent stronger joint feasibility screening before designation. If they subsequently show materially fewer late-stage failures on nitrogen, grid capacity or water than earlier developments, the claim that Dutch planning is learning recursive feasibility will be strengthened. If they stall in the same ways, it will be weakened.

P3 — Energy infrastructure. If national energy-infrastructure programming increasingly matches new electrification and industrial commitments to secured capacity before they are made, new debt of the unsecured-capacity form should decline. The growth or shrinkage of grid-related entries on the balance sheet would show whether it has.

P4 — Nitrogen architecture. If nitrogen governance acquires durable pathways, predictable long-horizon financing and transition arrangements that survive changes of government, the institutional part of this report's diagnosis will be weakened. If targets continue to be monitored while implementation arrangements are repeatedly reset, institutional continuity will be confirmed as a central explanatory variable.

Each prediction can fail. That is their purpose. If all four resolve in the Netherlands' favour, this report will have described a transition that succeeded, and the country will have shown that the wisdom it developed for water can govern its other constraints as well.

6.7 What the Balance Sheet Would Not Do

The balance sheet would not decide which commitments the Netherlands should make. It would not rank sectors, assign blame for past decisions or pass judgement on investments such as Lelystad. It would not prevent the government from committing on uncertain capacity when it judges the risk worth taking. And it would not resolve the distributional conflicts that make the hardest debts hard to repay; those require the legitimacy architecture described in Section 3.

Its purpose is narrower: to ensure that when the country borrows against its future feasibility, it does so knowingly, records the loan and plans its repayment.

6.8 How to Measure Success

The balance sheet's success should be judged over five years, by whether it changes how the Netherlands makes and tracks its commitments.

Five markers would show that it is working. The first balance sheet has been published and debated alongside the budget. At least one outstanding debt has been retired through a published pathway rather than a court ruling. Major new commitments routinely enter the balance sheet as contingent debts with repayment pathways, rather than appearing only when they fail. The total of outstanding debts has begun to fall, not merely to be reported. And public debate about housing, energy, aviation and agriculture has begun to ask not only what the country should commit to, but whether a feasible path to it has been secured.

That last marker matters most. The Netherlands became the country it is by refusing to accept that the sea set the limits of what it could build—and then by learning, after 1953 and again after the near-floods of the 1990s, to plan for a future it could not predict. The balance sheet would ask the same discipline of its commitments on land: not to build less, but to know, before it promises, whether the promise can be kept.

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7. Coda: The Makers of Land at the Limit of Promise

7.1 The Wealth That Matters

The Netherlands is rich in something few countries possess: a centuries-long, practical intimacy with limits. Its people have lived below the sea and beside great rivers for as long as they have been a nation. They have learned that the environment does not negotiate, that a dike is either high enough or it is not, and that the cost of being wrong is measured in lives. Out of that experience they built not only dikes and pumping stations, but institutions, habits of mind and methods of planning that the rest of the world now studies.

These are not small assets. They are the reason feasibility debt is a Dutch paradox rather than a Dutch condition. A country that did not know how to govern under uncertainty could be forgiven for borrowing against a future it did not understand. The Netherlands knows how. It wrote the method.

The wealth that matters in the coming decades is the ability to apply that method beyond the domain where it was learned. The constraints that now shape Dutch life—nitrogen and nature, grid capacity, water quality, space for housing and energy, the limits of an airport surrounded by homes—are as real as the sea and, in many respects, as uncertain. The Netherlands already knows how to govern such constraints. It has not yet chosen to govern all of them that way.

7.2 The Shift

The shift this report describes is not from ambition to caution. The Netherlands needs to build more, not less: more homes, more grid capacity, a transformed countryside, a sustainable energy system. The shift is in the order of operations.

Too often, outside water, the Dutch state has committed first and secured feasibility later. It authorised nitrogen emissions against ecological gains that had not yet occurred. It subsidised renewable projects before checking whether the grid could carry them. It allowed its national airport to operate ahead of the legal regime meant to govern it. It built an airport before the path to operating it was secure. In each case, the future was treated as if it had already arrived, and in each case, when it did not arrive on time, someone had to pay—farmers, developers, residents, public budgets, and the trust of citizens in the state's word.

The alternative is already Dutch. Adaptive delta management does not require knowing the future. It requires identifying the conditions under which a current path will fail, deciding early enough to build an alternative, keeping more than one path open, and revisiting the choice as evidence arrives. Applied beyond water, it

becomes a simple discipline: *test, then commit*. And where the test shows that a commitment rests on capacity not yet secured, record the debt, plan its repayment and settle in advance who will bear the loss if it must be repaid.

The cultural shift underneath is the one Room for the River already made. For centuries, Dutch *maakbaarheid* meant making land: forcing the environment to accommodate the country's plans. The newer wisdom is to make room: to accept a boundary, plan around it and allocate honestly what it costs. That is not a retreat from Dutch confidence in design. It is its maturation.

7.3 The Wider Significance

The Netherlands is not alone in borrowing against its future feasibility. Around the world, governments are making commitments—to decarbonise, electrify, build housing, restore nature—that depend on capacity not yet built, ecological recovery not yet achieved and legal regimes not yet enacted. Much of the energy transition in rich countries rests on grids that do not yet exist. Many climate targets rest on technologies not yet deployed at scale. Borrowing against future feasibility is becoming one of the defining risks of an era of simultaneous transitions.

What makes the Netherlands significant is that it experiences this risk earlier and more intensely than most, because its constraints are denser, its space scarcer and its courts more willing to enforce the commitments the state has made. And it is significant because it holds, in its own water governance, the most developed answer any country has produced. If the Netherlands can show that the method it developed for an uncertain sea can govern its other constraints—that recursive feasibility can be built into nitrogen, energy, housing and aviation—it will offer a model that every densely constrained society will need. If it cannot, the lesson for others will be sobering: that even the country that invented adaptive planning could not apply it where the losses were permanent and the politics divided.

The comparison with the reports that precede this one sharpens the point. Bhutan showed a country with a clear objective racing to keep new institutions answerable to it. Finland showed a competent system unable to transform quickly enough. The Netherlands shows something between the two: a competent system whose ambition sometimes outruns the feasibility it has secured, and whose remedy lies in a capability it already possesses.

7.4 The Honest Conclusion

This report has proposed a diagnosis, and it must be honest about what the evidence establishes and what it leaves open.

The evidence establishes that feasibility debt has recurred in several distinct Dutch domains: in nitrogen permitting, where future ecological capacity was credited in advance; in renewable energy, where subsidies were awarded before grid capacity was secured; and at Schiphol, where operations ran for roughly a decade ahead of the legal regime meant to authorise them. It establishes that the debt has been called in by different enforcers—courts, physics, finance—and that its repayment is hardest where the missing capacity is uncertain, the required losses permanent and the constraint tied to particular places. It establishes that the Netherlands already possesses, in adaptive delta management, a domestic architecture designed to prevent such debt, and that water's advantage lies in a complete package of continuity, independent integration, dedicated finance, alternative pathways and early decision points that other domains have acquired only in part. And it establishes that Dutch planning has begun, in several domains, to test feasibility before committing.

Much remains open. The evidence shows that feasibility debt recurs; it does not show that it explains most of what goes wrong in Dutch governance. Whether the recent shift toward earlier feasibility testing will reduce stalled and stranded commitments has not yet been demonstrated in outcomes. Whether water-quality obligations will produce a second shock after 2027 is unknown. Whether the Delta method's functions can be transferred to domains with different authorities, legal frameworks and losses is unproven. And whether Dutch politics can sustain the long-horizon arrangements needed to allocate permanent losses in agriculture—the arrangements whose abandonment left the nitrogen debt unpaid—is the hardest open question of all.

The transition is feasible. The Netherlands has already demonstrated, at home, every capability it requires. What it has not yet demonstrated is the will and the continuity to apply them beyond the domain where they were born.

7.5 A Final Word

In 1932, workers closed the last gap in the Afsluitdijk, the great dam that sealed off the Zuiderzee from the North Sea. Its builders knew that the most dangerous moment is the last: as the gaps narrow, the tide races through them, and a dam that is almost finished is not a dam at all. The final closures were made only when everything behind them was secure. Behind it, over the following decades, new land rose from the water. On some of that land, at Lelystad, stands a finished airport terminal that has not yet seen a commercial flight.

The two structures stand on the same reclaimed ground, and between them lies the argument of this report. The first was built by a country that understood that a commitment is only as strong as the path that secures it. The second was built by the same country, in an age of denser constraints and faster promises, before its path was secured.

The Netherlands does not need to rediscover how to govern under uncertainty. It taught the world how. It needs only to remember, on land, what it has never forgotten at sea: that the future cannot be spent before it is secured, and that the time to make room for it is before the commitment, not after.

The makers of land now stand at the limit of promise. What they do next will matter not only to the farmers, builders, residents and travellers whose futures depend on commitments being kept, but to every society learning, as the Netherlands already has, that the hardest constraint to govern is not the one outside the dike, but the one inside the decision.

Appendix A: Value Systems and Policy Mindsets — A Guide for the Dutch Context

A Note on This Appendix

The main body of this report speaks the language of feasibility, constraints and commitments. This appendix offers a complementary lens for readers who wish to understand the value-system dynamics beneath feasibility debt. It draws on Spiral Dynamics integral theory, simplified. It is optional, but it makes the report's underlying logic transparent.

A.1 The Basic Insight

Institutions and political cultures operate from different centres of gravity in how they understand governance, resources and change. These are underlying value systems—ways of constructing what feels real, legitimate and important—each a coherent response to particular life conditions, each with characteristic strengths and blind spots. What follows is a simplified map of those most relevant to the Netherlands.

A.2 The Value Systems in the Dutch Arena

Community and Place (sometimes called "Purple"). Rural Dutch life—the family farm passed down through generations, the village, the regional identity of Friesland, Brabant or the Achterhoek—carries a deep attachment to land and continuity. Strengths: stewardship, belonging, intergenerational commitment. Blind spot: permanent change to land use is experienced not as a policy adjustment but as a loss of identity, which is why the allocation of nitrogen losses became so much more than a technical question.

Order and Rule (sometimes called "Blue"). The Dutch constitutional order, independent courts, the rule of law and the European legal framework express a value system of binding obligations applied impartially. Strengths: predictability, protection of those with less power, and the capacity—demonstrated repeatedly in this report—to hold the state to its own commitments. Blind spot: a legal system can declare a boundary but cannot, by itself, allocate the losses that respecting it requires.

Achievement and Engineering (sometimes called "Orange"). *Maakbaarheid*, the Delta Works, Dutch agricultural productivity, Schiphol's hub strategy and the country's logistics economy express a value system of ambition, optimisation and technical mastery. Strengths: extraordinary accomplishment in engineering, trade and innovation. Blind spot: the confidence that feasibility can be made later—that the grid will be built, the ecology will recover, the law will be written—which is the cultural root of feasibility debt.

Consensus and Care (sometimes called "Green"). The polder model, environmental movements and the concern for nature, residents' health and future generations express a value system of inclusion, negotiation and protection. Strengths: broad ownership of decisions, sensitivity to those who bear costs, and the moral energy behind the court cases that enforced nature and climate commitments. Blind spot: a preference for continuing negotiation over imposing permanent losses, which can postpone allocation until a court or a crisis forces it.

Integrative and Adaptive (sometimes called "Yellow"). Adaptive delta management is one of the clearest institutional expressions of this mindset anywhere: planning that holds several futures in view, identifies tipping points and keeps alternative pathways open. Strength: governance under deep uncertainty. Blind spot: it has matured in one domain and has yet to be generalised.

A.3 Feasibility Debt as a Value-System Pattern

Feasibility debt arises where Orange ambition meets Green reluctance to impose permanent loss, within a Blue legal order that eventually enforces the boundary both have deferred. Orange commits; Green postpones the allocation of losses; Blue calls in the debt; Purple communities bear much of the repayment. Each value system acts reasonably from within its own logic. The pattern emerges from their interaction.

The Delta architecture shows the alternative: a Yellow integration in which Orange engineering ambition is disciplined by explicit uncertainty, Blue legal continuity protects long-horizon commitments, Green concerns are built into early decisions, and Purple communities are involved in allocating the losses—as Room for the River did. The transition proposed in this report is, in value-system terms, the generalisation of that integration beyond water.

Appendix B: International and Domestic Analogues

The proposals in this report draw on experience that does not transfer directly. The following are design analogues, not prescriptions.

B.1 Room for the River: Allocating Permanent Loss at Home

The most important analogue is Dutch. After the river floods and mass evacuations of the mid-1990s, the Netherlands adopted a new approach to river safety: instead of raising dikes indefinitely, give the rivers more space. The Room for the River programme, carried out between roughly 2006 and 2015, lowered floodplains, relocated dikes and created bypasses at more than thirty locations, in several places requiring the permanent relocation of farms and, in some cases, homes. It is widely regarded as a success. Its commonly cited lessons —a clear public purpose, early involvement of affected communities, fair and predictable compensation, local design freedom within a national framework, and programme stability across governments—are the conditions this report argues a legitimacy architecture for permanent loss must meet.

B.2 Adaptive Delta Management and Dynamic Adaptive Policy Pathways

The method at the heart of this report's counter-model was developed in the Netherlands and is now applied internationally, including in flood-risk planning for major river and coastal cities. Its core elements—adaptation tipping points, decision points set before them to allow for lead time, alternative pathways kept open, and periodic revalidation—define what recursive feasibility looks like in practice.

B.3 Australia's Murray–Darling Basin: Permanent Reallocation of a Scarce Resource

Australia's Murray–Darling Basin Plan sought to return water from irrigation to the environment in an over-allocated river system, partly by purchasing water entitlements from farmers. The experience shows both that permanent reallocation of a scarce resource is possible and how contested it becomes: buybacks provoked strong opposition in affected communities, and the balance between purchases and infrastructure investment has been repeatedly revised. It is a useful comparison for Dutch nitrogen policy, which faces a similar structure of permanent, place-specific adjustment.

B.4 Germany's Coal Commission: Negotiating a Compensated Exit

Germany's commission on growth, structural change and employment, established to plan the country's exit from coal, brought together industry, unions, environmental groups, regional representatives and scientists to agree a timetable and a package of regional support and compensation. Whatever its later difficulties, it

illustrates an approach to permanent, concentrated loss that settles the terms of exit, the compensation and the regional transition before the exit is forced.

B.5 Fiscal Risk Accounting

The Feasibility Balance Sheet borrows its logic from public finance. Governments increasingly publish not only their debt but their contingent liabilities—guarantees, obligations that may crystallise, fiscal risks from state enterprises—so that Parliament can see exposures before they become losses. The balance sheet proposed here applies the same principle to commitments made against physical, ecological and legal capacity.

Appendix C: The Governance as Engineering Connection

C.1 The Architectural Foundation

This report draws on the Governance as Engineering series, which models governance institutions as feedback control systems using control theory, information theory and cybernetics. This appendix summarises the parts that underpin the diagnosis and distinguishes what the papers establish from what this report infers.

As with the Bhutan report, this report inherits no prior estimate from the series' measurement programme. The Netherlands was not among the countries piloted in Paper VIII, and no Variety Gap figure is available for it.

C.2 The Papers in Brief

Paper II — Fractality as Stability shows that a controller operating on a single timescale cannot stabilise disturbances arriving on very different ones. Feasibility debt is, at root, a timescale problem: commitments made on political timescales against capacity that is built, recovered or legislated on much longer ones.

Paper XIII — Legitimacy treats legitimacy as an operational gain parameter. The Dutch rural transition is a case in which that parameter fell low enough that a technically coherent programme could not be sustained. Where repaying feasibility debt requires permanent losses, legitimacy becomes part of the constraint set.

C.3 Recursive Feasibility and Model Predictive Control

The report's central formal idea comes from optimisation and control engineering. In model predictive control, a controller chooses actions by optimising over a predicted future horizon, subject to constraints on the system's state. A well-known difficulty is that a controller can choose an action that satisfies every constraint now but leads the system into a state from which no feasible action exists at the next step. The property that guarantees this cannot happen is called **recursive feasibility**: every chosen action leaves at least one feasible continuation.

Feasibility debt is a violation of recursive feasibility in governance. The three forms documented in this report correspond to three ways of violating it:

- **borrowed future capacity** treats predicted future slack as present capacity, enlarging the apparent feasible set beyond the real one;
- **unsecured implementation capacity** verifies feasibility at one moment without securing it across the horizon, so that the feasible set shrinks before the action is completed;

- **anticipated legalisation** operates outside the current constraint set on the expectation that the constraint set itself will later move to include the chosen state.

The four-rung ladder—observe, forecast, conditionally preserve, revalidate and release—corresponds to increasing strengths of feasibility guarantee. The release rung reflects a trade-off familiar in constrained optimisation: tightening constraints to guarantee future feasibility reduces the performance achievable now. The optimal level of preservation balances the risk of future infeasibility against the cost of present exclusion.

C.4 What the Netherlands Adds to the Framework

The series has so far diagnosed deficits in execution, integration, feedback, synchronisation and calibration; ceilings of throughput and continuity; the compression of objectives; and the coupling of objectives to actuators. The Netherlands adds a temporal failure mode: a system whose individual decisions may each be feasible while its sequence of commitments is not. That suggests a candidate for future formal work: a paper on recursive feasibility in governance, modelling how commitment horizons, capacity lead times and enforcement mechanisms interact, and under what conditions a governance system can guarantee that its commitments remain executable.

Appendix D: Anticipated Objections

D.1 "The Netherlands is simply too full. The problem is lack of space."

Space is scarce, but scarce space is what the Netherlands has always governed. The relevant quantity is not hectares but land that remains usable once every constraint has been satisfied, and that depends on water, nature, grid capacity, drinking water and law. More importantly, the country already allocates scarce freshwater, grid capacity and nitrogen space through explicit systems. The difficulty this report identifies is not the existence of constraints but the practice, in several domains, of committing before they are secured.

D.2 "The courts caused the nitrogen crisis."

The courts did not create nitrogen deposition, the protected habitats it harms, or the European obligations to protect them. They enforced commitments the state had made in law and refused to accept uncertain future mitigation as present compliance. The permitting system they overturned had been built on capacity borrowed from the future. The courts called in the debt; they did not incur it.

D.3 "Judging Lelystad or PAS now is hindsight."

The report does not judge whether Lelystad should have been built or whether PAS was a reasonable attempt to break a deadlock. It records a sequence: commitment first, feasibility later. The purpose is not to assign blame for past decisions but to show a recurring structure and to propose instruments that make the risk visible before commitment rather than after.

D.4 "Requiring feasibility before commitment will slow down housing and the energy transition."

The report proposes neither certainty before commitment nor maximum reservation of capacity. It proposes that major commitments be accompanied by an assessment of which capacity is secured and which is borrowed, and that borrowed capacity be recorded and its repayment planned. Projects built on secured paths proceed; projects built on borrowed capacity too often stall halfway, as the paused infrastructure programme and the unconnected renewable projects show.

D.5 "Water is easier. You cannot compare it with nitrogen."

The report agrees that water governance has been most successful where options could be kept open and losses were temporary, and that its own hardest tests—permanent land-use change and a growing funding gap—are only now arriving. That is why it proposes transferring the functions of the Delta architecture rather

than its institutions, and why it adds a legitimacy architecture for permanent loss that the Delta model alone does not supply.

D.6 "*Gedogen* is a cultural stereotype."

The report does not claim that Dutch tolerance culture caused anticipatory enforcement at Schiphol, and it states that the evidence shows no such causation. It notes only that the long Dutch practice of formally tolerating the formally prohibited provides cultural resonance with an arrangement that might have seemed more exceptional elsewhere.

D.7 "Farmers are being blamed."

The report does the opposite. It identifies farmers—including those left without valid permits when PAS fell—as among those who bear the heaviest costs of repaying debts they did not incur alone. It argues that permanent losses must be allocated through fair, predictable and well-funded arrangements settled before they are forced, and it notes that when such arrangements were offered, many farmers chose them.

Appendix E: About this report and how it was made

How it was made

This report was generated by AI language models (Claude, ChatGPT, DeepSeek) from prompts by Björn Kenneth Holmström. Several models were prompted to analyse Dutch governance from different angles, applying the Governance as Engineering framework that he had set out. He chose the topic, posed the questions and decided to publish the result, but wrote little of the text himself.

The models were taken through the staged process developed for the South Korea and Bhutan reports, with its first step adapted to the case. Instead of an objective baseline, they began with a **constraint baseline**: mapping what binds Dutch development, how each constraint became binding and who enforces it. A constraint-interaction matrix followed, then a falsification matrix and two evidence passes. Several early hypotheses did not survive. The claim that the Netherlands lacks mechanisms for allocating scarcity was dropped when the models found explicit allocation systems for freshwater, grid capacity and nitrogen. An early reading that stalled projects are typically trapped by several constraints at once was weakened by project histories showing single and sequential bottlenecks—and then partly restored, at programme level, when a deliberate search for stalled rather than resolved projects corrected the sample's bias. The feasibility-debt hypothesis began with a single case and became the report's core only after independent cases were found in energy and aviation. A final comparison of institutional features across five domains was used to explain why water differs. The surviving claims were recorded in a claim-and-evidence ledger with wording ceilings. All of these steps, the evidence they cite and the conclusions drawn were produced by the models; whether the final text stays within the ledger has not been independently checked.

The report has not been systematically fact-checked. Statements about Dutch institutions, laws, projects, figures and sources may be inaccurate or invented, and should be verified against primary sources before being relied on or cited. Where the report says "evidence shows", that is the models' characterisation of external sources, not a verified finding; where it uses the language of the framework, it refers to synthesis within Governance as Engineering. The report registers predictions that could weaken or overturn its diagnosis. Like any work of synthesis, it is provisional.

Perspective and limits

Neither the models nor the person who prompted them write from within Dutch institutional life. Björn Kenneth Holmström is not Dutch, does not live in the Netherlands and has no lived experience of Dutch politics, its farming communities or its planning institutions. He has studied engineering physics and has worked with mathematical optimisation at TOMLAB Optimization; the report's central concept, recursive feasibility, comes from constrained optimisation and control theory. The framework applied here also draws

on complexity science, developmental psychology and governance theory. Much of the texture of Dutch governance is inaccessible from outside, and Dutch readers will see what this report misses. Feedback, criticism and corrections are welcome.

The report refers to related material on this website—the Global Governance Frameworks, the Governance as Engineering working papers and the Country Reports for Systemic Change—which were produced in the same way.

The Country Reports Series

This report is the nineteenth in the series of Country Reports for Systemic Change, all produced in the same way. The earlier reports examined Germany through the lens of an execution deficit, France through an integration deficit, Sweden through a feedback deficit, India through a synchronisation deficit, the European Union through a coherence deficit, the United Kingdom through a control-delivery deficit, Brazil through an accumulation deficit, Russia through a power-vertical deficit, the United States through an integration deficit, Finland through a throughput constraint, China through a calibration deficit, Japan through a continuity trap, Nigeria through a substrate deficit, Israel through a boundary deficit, Spain through a transition trap, Switzerland through a boundary-expansion deficit, South Korea through a tournament trap, and Bhutan through an interface race.

Together, the reports propose diagnoses spanning first-order deficits of execution and integration, second-order constraints of velocity and paradigm lock-in, the foundational challenge of substrate construction, constitutional incompleteness, the compression of objectives and the coupling of objectives to institutions. In the Netherlands the report sees a temporal failure mode: a system whose individual decisions may each be feasible while its sequence of commitments is not. The series frames the Netherlands as the makers of land at the limit of promise, and asks whether a country that taught the world to plan for an uncertain future can apply that wisdom to its own.

Appendix F: Case File — Nitrogen and the Renewable Subsidy Cohort

F.1 Nitrogen: From PAS to the 2030 Target

2015. The Programmatic Approach to Nitrogen (PAS) comes into force, combining source reduction, nature restoration and a permitting framework that allows new activities to be authorised partly on the basis of ecological gains expected from future measures. Smaller activities may proceed through notification rather than full permits.

May 2019. The Council of State rules that PAS cannot serve as a permitting basis. Under the Habitats Directive, as interpreted by the EU Court of Justice, authorities must be convinced that site integrity will not be adversely affected, with no reasonable scientific doubt remaining; expected future gains do not meet that standard. The permission basis for a wide range of activities falls. Activities that relied on notification—the PAS reporters—are left without valid permits.

2019 onward. A nitrogen registration system is created to allocate freed deposition space, originally with up to 70 percent available for development and at least 30 percent reserved for nature, and with priority for housing and selected infrastructure. Priorities are later adjusted to favour the legalisation of PAS reporters.

2022–2024. A national programme for rural areas is proposed, with provincial area programmes and around €24.3 billion reserved, partly through a dedicated transition fund. Farmer protests intensify; the Farmer–Citizen Movement wins the March 2023 provincial elections; a broad agricultural agreement fails in June 2023. In 2024 the programme is ended, and in November 2024 the fund legislation is withdrawn, with €20.5 billion redeployed.

December 2024. The Council of State tightens the treatment of internal offsetting, with immediate effect on pending and future procedures.

January 2025. The Hague District Court orders the state to meet its statutory 2030 target of bringing half of nitrogen-sensitive protected habitat below its critical threshold, with priority for the most overloaded areas and a €10 million penalty if the target is missed. The judgment remains executable pending the state's appeal.

2026. New deposition space from closed pig farms becomes available but is described as limited, because an additionality test reserves much of the reduction for ecological recovery.

F.2 The Renewable Subsidy Cohort

Before autumn 2019. The national subsidy scheme for renewable electricity awards support without a project-specific check that grid transport capacity exists.

Mid-2019. The government acknowledges that the grid is under heavy pressure from existing renewable projects and that further awards in locations without capacity would worsen the problem.

Autumn 2019. A transport indication from the network operator becomes a condition of award.

December 2019. Parliament records around 700 megawatts of subsidised projects unable to obtain connection or transport capacity.

Spring 2020. Network operators have issued more than 9,000 transport indications.

2026. The government acknowledges that many solar projects with a transport indication are later unable to connect, because indicated capacity is not reserved through development, and investigates how to make the indication reflect future availability more reliably.

F.3 What the Cases Show

Both cases show commitment preceding durable feasibility. They differ in their enforcer—a court for nitrogen, physics and network operators for the grid—and in the difficulty of repayment, which is far greater for nitrogen because its missing capacity is uncertain, its repayment often permanent and its constraints tied to specific places.

Appendix G: Case File — Schiphol and Lelystad

G.1 Schiphol's Anticipatory Enforcement

2015. A new system of runway use, the New Norms and Enforcement System, is applied at Schiphol to reduce overall noise nuisance. Pending its incorporation into the legally binding airport decree, the aviation inspectorate is instructed not to sanction exceedances of the existing legal noise limits that result from following the new system.

2021. The inspectorate warns the minister that the arrangement has become legally vulnerable and that timely legalisation is not in prospect.

2022. The government decides in principle to end anticipatory enforcement.

March 2024. The Hague District Court finds that the state has failed adequately to protect residents and has insufficiently enforced existing rules, and must restore effective legal protection.

July 2024. The Supreme Court rules that measures reducing airport operating capacity must follow the EU's balanced-approach procedure.

2025. A partial revision of the airport decree introduces annual limits of 478,000 flights and 27,000 night flights, but does not end anticipatory enforcement.

September 2026. The Council of State publishes its advice on the new airport decree intended finally to end the arrangement and restore residents' legal position.

The arrangement is specific to Schiphol. The evidence does not show that anticipatory enforcement is a general instrument of Dutch administration.

G.2 Lelystad Airport

After 2015. Following the airport decision, the runway is lengthened and widened, the control tower enlarged and the terminal completed, with commercial opening prepared for April 2018.

End of 2016. Opening is postponed because the required airspace and route arrangements are not ready.

2019. Following the PAS ruling, a nature permit is required for the intended commercial operations.

2019–2026. The low-altitude route issue remains contested; by 2026 routes allow continuous climbs and descents and the disputed low-level segment has been removed.

April 2026. The government sets an ambition for commercial operations from October 2027, initially around 10,000 flights a year, conditional on a nature permit, an amended airport decree and completed operational preparations.

May 2026. Parliamentary papers record no clear prospect of an irrevocable nature permit.

G.3 What the Cases Show

Schiphol shows the anticipated-legalisation form of feasibility debt, and how borrowed space acquires a constituency the longer it lasts. Lelystad shows unsecured implementation capacity in a long-lived project whose binding constraints changed as the years passed. Neither case file passes judgement on the merits of the underlying decisions; both record the sequence in which commitment preceded durable feasibility.